MOMENT-METHOD ANALYSIS OF A NON-UNIFORM MONOPOLE ANTENNAS

BY

OLUBUNMI ADEOLU ADENEKAN

(980403013)

UNDER THE SUPERVISION OF

DR. A. A. AYORINDE

A MAIN RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF MASTER OF SCIENCE (MSc) DEGREE IN ELECTRICAL & ELECTRONICS ENGINEERING AT THE UNIVERSITY OF LAGOS

MARCH 2023

CERTIFICATION

This is to certify that this research work was carried out by **OLUBUNMI ADENEKAN** with matriculation number **980403013** under the supervision of **DR. A. A. AYORINDE**.

OLUBUNMI A. ADENEKAN ……..……..

(Project Student) **Date and Signature**

Dr. A.A. AYORINDE ……..……..

(Project Supervisor) **Date and Signature**

Dr. A.S. ALAYANDE ……..……..

(PG Coordinator) **Date and Signature**

Prof. T. O. AKINBULIRE ……..……..

(Head of Department) **Date and Signature**

DEDICATION

This project is dedicated to Almighty God for His undeserved mercy and grace as well as to my wife, my children and family for their encoragement and support towards the completion of this project.

ACKNOWLEDGEMENT

Firstly, I am sincerely grateful to God Almighty who has been my help in the past, present and will be in years to come , for His love, passion over my life and the lives of my family members and loved ones. I am very grateful to God for giving me the mental and physical strength to undertake this research work.

The success and final outcome of this project required a lot of guidance and assistance from my supervisor, Dr. A. A. Ayorinde and I am extremely priviledged to have gotten this support during the course of the research work.

I also want to specially thank my wife, Mrs. Mojisola Adenekan, FCA for her support and continuous encouragement to ensure the completion of this project irrespective of my very tight schedule. I will aslo like to thank the immediate past Head Of Department, Elect/Elect Engineering in person of Dr. S. Adetona for his valuable contribution and constant encouragement towards the succesful completion of this project.

My heart felt gratitude also goes to my parent, Mr. D. O. Adenekan and Late Mrs Florence Adenekan for their love, concern, encouragement and more importantly their prayers that saw me through the university and made this project a success.

Finally. I say a hearty thank you to all my MSc colleagues, my colleagues in the office as well as my family and friends that contributed to the succesful completion of my MSc programme. Thank you all.

ABSTRACT

In wireless communication systems, antennas play such a critical role that without them communication will be impossible.

The radiation problem of a non-uniform monopole antennas singly or multiply exicted by delta-gap generators, reactively and resistively loaded at discrete points is addressed in this dissertation using the method of moments in the MATLAB environment. The antenna is modelled as thin-wire which constrained the current to flow only along the antenna's axis which validated the vector potential integral for solution of the thin-wire problem.

Upon the determination of the current distribution on the antenna input impedances and admittances as well as the radiated fields of the monopole antenna computed as graphically illustrated. It is shown that when the non-uniform monopole antenna is reactively loaded, the current distribution increases as frequency increases and the field patterns take the shape of figure 8. However, when it is resistively loaded, the current distribution increases as well as the frequency and the field patterns take the shape of figure 8 at first then deform into other shapes subsequently.

It was also observed that multiple excitations yield greater propagation distance ,which is a measure of directivity while single excitation resulted into more angular coverage for the straight monopole antenna.

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CHAPTER ONE

INTRODUCTION

1.1. Background

Antennas are the main components used in electronic devices and systems for the acquisition of signals and the dissemination of information. They play a very central role, especially in wireless communication systems, in determining the quality of signals acquired by receivers and the quality of signals transmitted. In deep space communication and exploration, high-gain antennas are used by National Aeronautics and Space Administration (NASA) to communicate and control space probes that are billion of miles away from earth. It is obvious from the foregoing that antennas are a dominant feature in human advancement in communication. An antenna is conceptually the interface between a radio system and the external environment in which the radio operates [1]. It converts information signals into electromagnetic (EM) waves and vice versa. Due to the unique strengths and weaknesses of antennas, there are different types of antennas that are well-suited for specific application domains.

Monopole antennas are mostly used in airborne communication systems areas because of their wide frequency range requirements [2],[46]. In applications that require antennas that have omnidirectional coverage, compact size, and excellent radiation, dipole antennas are the choice antennas [3]. When an installation is characterized by limited space and ultra-high frequency requirements, the rectangular antenna type comes into play as the antenna of choice [4],[5]. There are instances in which an antenna is expected to have a wide array of functions; circular loop antennas fit into these scenarios because they have a known range of different use cases [6] ,[7].

Despite their ubiquitous nature in wireless communication systems and specialized domain of applications, the design of every type of antenna has its unique challenges, and the variation in the design challenges is proportional to the sub-specialization of the antenna in the domain of application.

Wire antennas are the most familiar type of antennas because they are seen virtually everywhere - on automobiles, buildings, ships, aircraft, etc. It is the oldest, simplest, and cheapest form of antennas.

In [8], improved user – oriented computer programs were presented and described for analyzing the electromagnetic behaviour of arbitrarily bent thin -wire antennas and scatterers.

A given problem can involve several wires of different shapes and radii and the wires can be excited or loaded at arbitrary points along their lengths. Matrix methods are then used with method of moments to compute various quantities of engineering interest.

For radiation problems, the current distributions on the wires are determined along with appropriate near and far field patterns and input admittances corresponding to the feed points.

In this research work, the current distributions and the corresponding radiation patterns on a straight monopole antenna are studied comprehensively.

1.2. Motivation

Recently, numerous studies have been conducted on analyzing the electromagnetic behaviour of arbitrary thin-wire antenna using method of moments. [8]. The FORTRAN programs was used for implementation and computations performed using the matrix suggested by Harrington (1967). The learning curve for the FORTRAN is steeper and the programs only tested reactive load. Hence, the desire to study the effects of different types of loads on the current distributions and radiation patterns on monopole antenna using MATLAB program which already has tons of inbuilt matrices functions and toolboxes is pursued here.

While the existing FORTRAN program only tested reactive load, this project seeks to investigate the current distributions and radiation patterns of the same wire antenna geometry to loading condition, resistive load, and when the antenna is resistively and reactively loaded.

1.3. Aim and Objectives

The aim of this research work is to analyze the electromagnetic behaviour of monopole antenna with single and multiple excitations using method of moments.

Therefore, the following objectives are proposed:

1. To determine the input impedances and admittances at the points of excitation on the monopole antenna.
2. To determine the current distribution along the length of the wire antenna.
3. To compute the far-field zone radiation patterns.

1.4. Problem Statement

Wire antennas are the most versatile antennas in application. They are used in diverse communication process, ranging from TV, FM Broadcast, satellite, radar and other forms of wireless communication. There are different forms of wire antenna and they are categorized according to the geometry of the linear structure. Some of which include; the dipoles, monopoles, arbitrary bent wires, folded dipoles, linear arrays etc. [9]

A given problem geometry can involve more than one wire and it is not necessary that all the wires have the same shape or even the same radius. The wires can be loaded discretely at arbitrary points and for radiation problems they can be excited or fed at any arbitrary point or points along their lengths.[8]

1.5. Methodology

In the sequel report [8], the FORTRAN programs were written and executed for a given problem which can involve several wires of different shapes and radii. The wires can be excited or loaded at arbitrary points along their lengths.

In this dissertation, the electromagnetic behaviour of monopole antenna ia analyzed using the method of moments. The existing FORTRAN program is converted to a MATLAB program which already has tons of inbuilt matrices functions and toolboxes. In applying the method, each wire in the problem geometry is thought of as a number of short subsections or segments

connected together. It is assumed that a minimum of twenty segments should be used per wavelength and each segment should be at least five times the radius of the wire

1.6. Scope of Work

The scope of this reaerch work is confined to the method of moments analysis of a monopole antenna with single and multiple excitations.

1.7. Dissertation Outline

The report is organized in the following order:

Chapter one gives the introduction to the whole work. It gives the background study, problem statement, aims and objectives, scope of the work and the dissertation outline.

The chapter two of this thesis presents a literature review on fundamental parameters of antenna such as radiation patterns, current distributions and impedance proprties.

In chapter three, the details analysis of method of moments are presented and its application to the evaluation of the performance of an antenna. The chapter also looks at determination of the generalized impedance matrix, radiation patterns and current distributions of staright wire antennas.

The implementation tools, MATLAB version 9.8(R202a) was used in implementing the algorithm.

The simulation output results and discussions are presented in chapter four, and in chapter five, the conclusions of the dissertation and future research are presented.

CHAPTER TWO

LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1. Introduction to Antennas

In wireless communication systems, the antenna is one of the most critical components.

A good design of the antenna can relax system requirements and improve overall system performance. A typical example is TV for which the overall broadcast reception can be improved by utilizing a high performance antenna. An antenna is the system component that is designed to radiate or receive electromagnetic waves. In other words, the antenna is the electromagnetic transducer which is used to convert, in the transmitting mode, guided waves within a transmission line to radiated free-space waves or to convert, in the receiving mode, free-space waves to guided waves [10]. In a modern wireless system, the antenna must also act as a directional device to optimize or accentuate the transmitted or received energy in some directions while suppressing it in others [10].

There are different types and forms of antennas, each applicable or used in various systems. Some of the various types of antennas are the wire antennas, aperture antennas, microstrip antennas, yagi-uda array antennas, reflector antennas, lens antennas, etc. [10]. The concept of antennas and their applications in wireless communications have been an active area of research, with many innovative ideas in design and fabrication being churned out of industry on a regular basis. As a result of this, literature abounds with studies on thin-wire antennas, their design methodology, and the techniques for the analysis of their performance.

In electromagnetic systems, method of moments is an established mathematical tool for the analysis of antenna radiation problems.

Antennas serve this purpose because they are electromagnetic devices whose performance is strongly governed by radiation patterns and behaviour.

Therefore, this literature review will also cover Method of Moments technique and establish a link as to how it relates to the analysis of the performance of antennas.

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2.2. Theoretical Framework

In analysing an antenna system, the sources of excitation are specified, and the objective is to find the electric and magnetic fields radiated by the elements. Once this is accomplished, a number of parameters and figures-of-merit that characterize the performance of the antenna system can be found. When designing an antenna system, the characteristics of performance are specified, and the sources to satisfy the requirements are sought [11].

Based on the cylindrical symmetry of the wire antennas, its analysis is carried out in the cylindrical coordinate especially in determining the fields near the antenna. At far distances, the radiated fields are best described in spherical coordinate because any finite current source appears as a point source from far distances [11].

For this reason, It is necessary to note that as point sources, the angular distribution of the radiation pattern is independent of the linear distance [9].

2.3. Fundamental Parameters of Antenna

When describing the perfomance of an antenna, definitions of various parameters are necessary. Some of the parameters are interrelated and not all of them need be specified for complete description of the antenna performance.

Parameter definitions will be given in the section below as related to thin wire antennas [10].

2.3.1. Radiation Pattern

One of the most fundamental properties of an antenna is its radiation pattern, as many of the antennas performance parameters pertain to various aspects of the pattern [12].

An antenna radiation pattern can be defined as "a mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates[12].

In most cases, the radiation pattern is determined in the far region and is represented as a function of the directional coordinates. Radiation properties includes power flux density, radiation intensity, field strength, directivity, phase or polarization" [11].

An antenna pattern is three dimensional, and therefore its description requires a three dimensional coordinate system. The spherical coordinate is a more appropriate coordinate system to use because the radiation patterns may be expressed in terms of the electric field intensity, for example at some fixed distance 'r' from the antenna, at all points on the spherical surface at that distance. If an antenna is imagined to be located at the centre of a spherical

coordinate system, its radiation pattern is determined by measuring the electric field intensity over the surface of a sphere at some fixed distance [12].

The complementary problem of radiation pattern was discussed in [13]. By using a variational formulation, the author obtained for large screens, simple expressions for the change of the radiation pattern as a function of the diameter of the screen. There was no restriction imposed on the length of the antenna. The most important difference between the pattern of an antenna over a finite screen and an antenna over an infinite screen is the presence of lobes over the antenna. However, while it is possible to obtain theoretically the entire pattern over a finite screen, experimental comparisons are in reality limited to the region of small angles where the lobes occur [13].

A geometrical theory of diffraction was used to solve the problem of scattering by a finite, perfectly conducting right circular cone that was excited by a quarter- wave monopole protruding from the centre of its base [14]. The radiation patterns for the monopole excited cone was computed. Three model cones having semi- angels of 10, 15 and 20 degrees were fabricated in brass and subjected to radiation measurements in a microwave room. For the largest cone, there was an agreement between the theory and experiment unlike the medium and the smallest cones.

The problem of calculating the radiation pattern and input impedance of a monopole antenna centred on a circular ground plane has existed for many years. Since it is known that the radiation pattern of an antenna is not critically dependent on knowing an accurate distribution of electric currents over its surface, even for large surfaces such as paraboloids, attention has therefore been turned to finding radiation pattern from such physically real estimated currents on circular ground planes [15]. The method is a unified one for both calculations of ground plane radii of 0.6λ and 3λ and much preferred to the edge current method for the calculation of radiation patterns, particularly those of small ground plane radii. The fields diffracted by a body made up of finite axially symmetric cone frustums are obtained using the concepts of the geometrical theory of diffraction.

Charles and Leon [16] used the wedge diffraction coefficients to determine an equivalent current at the edge of a discontinuity such as the edge of the planar aperture, disk or the edge of the wedge formed by a ring slope discontinuity. These equivalents currents are then integrated using the radiation integral to determine the scattered fields. This solution is readily applicable to both antenna and scattering problems.

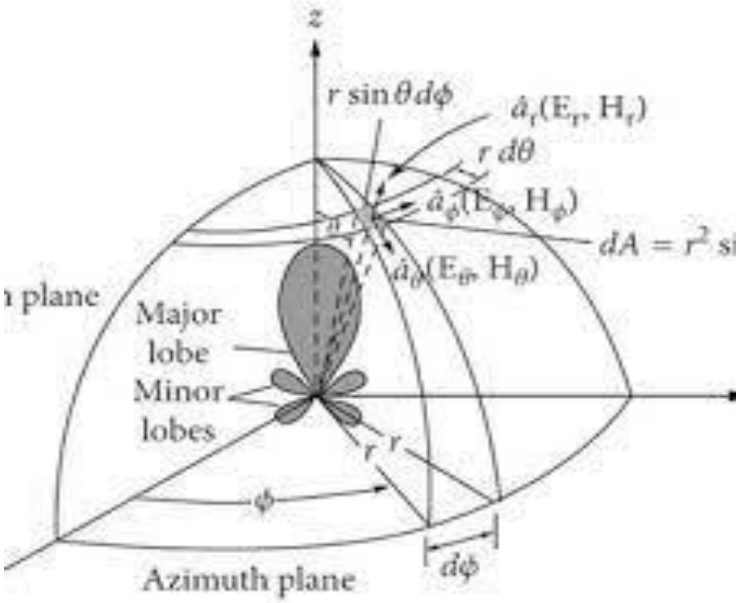


Figure 2-1: Coordinate system for antenna analysis

Alfred [17] discussed in his paper, how the geometrical theory of diffraction is applied to calculate the radiation pattern and impedance of a monopole antenna on a perfectly conducting circular ground plane of limited extent. The purpose of the paper is to demonstrate the usefulness of the geometrical theory of diffraction in the solution of a class of radiation problems, which arise often in practice. The pattern calculation consists of a superposition of a ray emanating from the monopole itself and two rays emanating from the edge of the circular ground plane. The radiation pattern of the disk-monopole antenna will then be calculated and compared to an existing result from the literature. It was observed that there was a good agreement between the measured and calculated pattern, especially in the periodicity of the interfering signals and also in the pattern amplitude near θ = 90º.

However, in the θ = 0º and θ = 180º directions, the calculated signal is significantly lower than the measured value [17].

The field pattern of an asymmetrically driven long antenna is obtained by the Wiener-Hopf procedure [18]. A dipole antenna of several wavelengths long with multiple loadings or excitations exhibits drastically different characteristics as the positions or the values of the loadings are varied. The loadings mentioned may either be passive or active. The purpose of the study is to obtain a simple algebraic expression for the field pattern of an asymmetrically driven antenna with a single generator so that the field pattern of an antenna with multiple

loadings may be found by superposition. This is accomplished by a procedure that makes use of the Wiener-Hopf integral equation.

One of the advantages of the present long antenna theory is that the complicated problem of mutual coupling effects between elements of an array is automatically taken into account and furthermore, a simple and accurate formula for a field pattern of a long antenna can be made available [18].

2.3.2. Current Distribution

The current distribution on the wire antenna or scatterer is a particular case of the general boundary value problem (BVP) involving conducting bodies in a known impressed field. The incident field or source voltage induces current on the antenna. The current in turn generates fields that radiates and propagates into space. The total field becomes the sum of the incident and the radiated fields which must varnish at the boundary of a perfect conductor. This boundary condition is necessary to determine the current distribution induced on the antenna. With knowledge of the current distribution on the antenna surface, all other parameters of engineering interest can be obtained by numerically evaluating the formulas [9].

A simple and qualitatively accurate representation of the current distribution of a dipole antenna is derived in [19]. The purpose of the paper is to give, for practical reasons, a theory of the dipole antenna which can give a simple yet accurate algebraic expression for the current distribution everywhere on the antenna even when antennas are not short or are asymmetrically driven. An outgoing travelling wave of current is generated along the dipole antenna when it is driven by a time-harmonic source. It travels along the two arms of the dipole with a speed almost equal to the speed of light, and decays slowly as a result of radiation. The current distribution on the antenna is just the result of the superposition of the outgoing current wave and all the reflected wave.

The characteristics of thin -wire , centre-fed , linear antennas have been studied extensively by R. W. P. King [20]. Through the use of an integral equation technique leading to the current distribution along such an antenna.

A travelling-wave linear antenna supports a distribution of current which is essentially an outward travelling wave. As compared with a conventional dipole, the travelling wave antenna has two practical aspects that are desirable for certain purposes: a broadband character whereby the input impedance is relatively independent of frequency, and a modified radiation pattern having a wider beamwidth and a notable absence of minor lobes for a long antenna. The

objective of the study was to investigate the possibility of realizing a traveling-wave antenna through the use of reactance loading technique due to dissipation in the resistive loading.

The current distribution along the antenna is obtained approximately in terms of its dimensions, the excitation frequency, and the impedance and position of the double loading [20].

The problem of determining the distribution of current in and the impedance of a cylindrical antenna asymmetrically driven by a discontinuity of scalar potential is formulated in [21].

If the center-driven cylindrical antenna is symmetrical with respect to its center, the currents in the halves are the same, I(-z)=I(z), and a single integral equation is sufficient for analysis.

However, if the antenna is not center- driven, numerous complications arise. The only manner in which an asymmetrical antenna can be driven without introducing a radiating transmission line is by a generator completely within the metal surfaces of the antenna [21].

The first work on traveling-wave dipoles was reported by Altschuler [22], who inserted lumped resistors at a quarter wavelength from the ends of the antenna. Although this location of the resistors is not critical, the traveling-wave nature of the current diminishes as the frequency is changed so that the lumped resistors are no longer at the maximum of the current.

The distribution of current on the end quarter-wavelength of the loaded antenna behaves very much like that on the un-loaded antenna. An approximate solution is described for the current distribution on a long, thin cylindrical antenna driven at an arbitrary point along its length by David in [23].

The electromagnetic behaviour of a cylindrical antenna can be advantageously modified by the insertion of lumped active or passive elements at convenient points along its length.The purpose of this communication is to describe a supplemental theory that is useful for both the synthesis and analysis of multiply loaded or multiply driven antennas. The solution sets no fundamental limitation on the electrical length of the antenna that is considered, and it yields a tractable description of the current distribution, a requirement for studies of antenna synthesis. Properties of a center-driven cylindrical antenna, characterized by a pure traveling wave of current was investigated in [24].

A solution is obtained specifically when the current is represented by an outward traveling wave with no reflected wave. It is proposed to investigate the cylindrical antenna with a variable internal impedance per unit length. In particular, it is desired to determine an axial

distribution of the internal impedance for which a pure outward traveling-wave exists on antenna of finite length. A theoretical study of the long dipole antenna was carried out by Tai[25], using an approximate procedure that makes use of the Wiener- Hopf integral equation.

However, as at that time, experimental evidence was inadequate for any meaningful comparison between theory and experiment. For this reason, no effort was made to improve the theoretical formula of the current distribution on the antenna which was known to be inaccurate near both the driving point and the ends.

Liang [17], did modification and revaluation in the original work of Wu. The modification introduces virtually no complication in the sense that the formulas for the current distribution and the input admittance are still algebraic and involve no integration. Most analyses of the distribution of current and impedances of cylindrical antennas yield accurate results only for antennas that are not much longer than about two wave-lengths [18].

However, an important exception is the work of Wu [25] which utilizes the Wiener-Hopf technique to analyze very long antennas [26]. However, no simple formula for the current is given. For some applications, such as problems in superposition, a simple trigonometric representation of the current and the corresponding driving-point impedance of a very long antenna is useful.

In his paper, Jebli [27] considered the electromagnetic field integral equation (EFIE) for computing the total current distributions on antenna surfaces, and studied the two standard methods by comparing their solution. These two methods are the direct method of moments developed by Harrington [28] and the iterative conjugate gradient method (CGM). The comparative study was done at the level of numerical solutions, achieved in terms of computational advantages and disadvantages as well as their ability to determine the current distributions on metal surfaces.

Electromagnetic field integral equation has been successfully used in the computation of finding the current distribution over the surface. Solving for current distributions on metal surface usually incorporates formulation of Electric Field Integral Equation [28] Few geometries yield solutions (analytical, numerical approximation) have been successfully used. But mostly used method is the method of moment. By his method , the initial operator equation can be formulated by employing boundary condition for tangential incident **(E^{in})** and impressed **(E^{sc})** electric field;

$$\mathbf{E}^{in} + \mathbf{E}^{sc} = 0 \quad (2.1)$$

Scattering or radiation problems are essentially identical, the only difference is that the 'incident" field for the driven antenna is the applied electric field in the feed. The total electric field is a combination of the incident field and the scattered field.

$$\mathbf{E}^{tan} = \mathbf{E}^{in} + \mathbf{E}^{sc} \quad (2.2)$$

The distinction between antennas and scatteres is primarily that of the location of the source. If the source is on the object it is considered an antenna: if the source is distant from the object it is viewed as a scatterer. It was concluded in this paper, that the direct Method Of Moment is more efficient than other iterative Cojugate Gradient Method, based on its performances, the number of iterations needed for convergence, storage, and the level of precision [29].

The feasibility of direct numerical calculations of antenna integral equations is investigated [30]. In the study of antenna theory, a knowledge of the current distribution is of fundamental importance. Such data may be obtained either by measurement or by solving the antenna integral equation. Integral equations are difficult to solve even for the simplest case of a dipole antenna. The purpose of this paper [30], is to present an investigation of the feasibility of direct numerical calculations of antenna integral equations. To simplify the discussion, the trapezoidal rule of integration is assumed throughout, although it is realized that in a practicalcalculation, better integration schemes, such as quadratic rule, etc, may be useful.

2.3.3. Impedance Properties of Antenna

The input impedance of an antenna is a characteristic of considerable interest to Engineers since they are concerned with the goal of supplying the antenna with the maximum amount of transmitter power available from the antenna. For linear antennas that are relatively small in size, it is possible to make some reasonably good estimates as to the magnitude of the input resistance. In most practical cases, the input impedance of even a simple antenna is affected to a considerable degree by the terminal conditions at the point where the transmission line feeds the radiator [31].

Meier and Summers [32] made an investigation to obtain some fundamental information concerning the relation of the impedance of a vertical antenna over a finite ground plane as a function of the size and shape of the ground plane when dimensions of the ground plane are relatively small in terms of wavelength. Antenna impedance measurements were made on

circular and square ground plane varying in dimensions from 1.7 to 6.1 wavelengths. The results indicate comparable amplitudes of oscillation for both 'thick' and 'thin' antennas and a noticeable damping oscillation as the ground plane increases in size. Rather large amplitudes of variations are evident which reveal that the antenna impedance is very critical with respect to the ground-plane dimensions. The amplitudes of oscillation of the square ground planes are considerably less than those of the circular ground planes , and the damping of oscillation is more rapid.

Harry [33], is concerned with the determination of the driving-point impedance of a monopole which is located at the center of the base of a perfectly conducting, right circular cone. The cone is assumed to be sufficiently large that it is valid to apply ray-optical theory. It is shown that the field which is reflected back to the monopole from the edge of the base may be considered as arising from an equivalent magnetic current located at the base edge, the magnitude of which is determined by wedge-diffraction principles. To find the monopole input impedance, Harry [33], used what is essentially a perturbation procedure to deduce a correction term to be added to the impedance of the same monopole but on an infinite ground plane. The method is based on ray-optical theory, which is used first to obtain an equivalent magnetic ring current with which to replace the edge of the base. The change in monopole impedance then follows directly by use of the induced EMF method.

A technique for computing the input impedance of a monopole on a finite ground plane is provided [34]. In this paper, the principle of the magnetic ring current used for the circular ground plane is extended to the polygonal case. The method converges to the circular case result with increase in the number of sides. The technique can be summarized in three steps. First, an impedance matrix is formed for the monopole on an infinite ground plane. Second, an impedance component determined by the diffraction effect at the edge of the ground plane is added to the element of this matrix. Third, by inverting the resultant impedance matrix, the input admittance is determined.

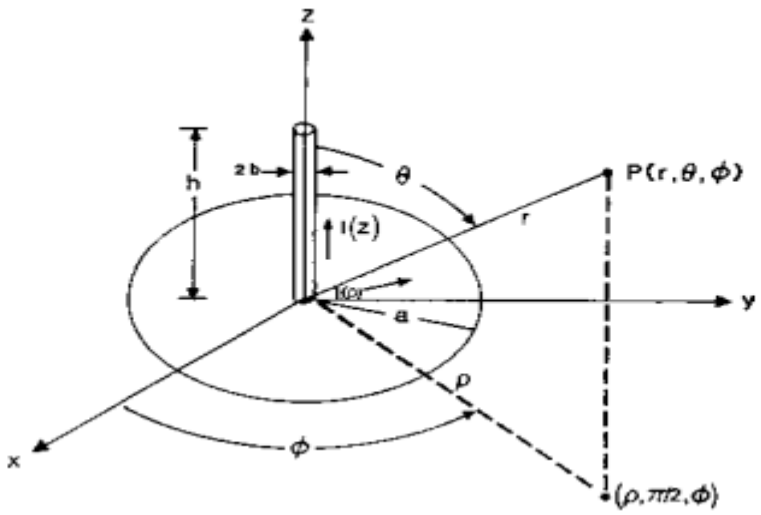


Figure 2-2: Monopole antenna at the centre of a circular ground plane

James [35], on the other hand, derived a simple formula for the change of the input impedance of an antenna as a function of the diameter of the ground plane. This paper treats the problem of a monopole antenna erected vertically in the center of a circular screen. An integral equation is obtained for the electric field in the plane of the screen and outside the screen. Meier and Summers [32] measured experimentally some effects produced by using a finite screen. Their results indicated that measurements taken over a finite screen may differ appreciably from those taken using an infinite screen. However, these results are not of prime interest but what the focused was is to know whether the screen is large enough so that for practical purposes it may be considered an infinite screen. The formula for the impedance change is given below [32]:

$$z - z_0 = \frac{9.95}{(d/\lambda)} e^{2\pi i(d/\lambda)} \left| k \int_0^h \frac{I(z)}{I(0)} dz \right|^2 \; ohms \tag{2.3}$$

The problem of a vertical monopole situated over a circular perfectly conducting screen lying on a finitely conducting ground is considered by James and Surtees [36].

Usually the most important factor contributing to the inefficiency of low-frequency antennas are seems to be in the ground system. The rules for ground system design are usually empirical and based on results of experiments on existing installations.

An important related problem is the actual change of input impedance of the antenna for different sizes of ground systems. The analysis has been carried out for a vertical antenna situated centrally over a perfectly conducting disk by Leitner and Spence [49] and more recently by Storer [35]. But, However, they only examined the case when the surrounding medium was free space. A more appropriate situation is when the disk is lying on the surface of a homogenous conducting half- space which corresponds to the ground. It is not surprising that the nature of this problem is in general very difficult.

The possibility of making acceptably accurate input impedance and gain measurements of monopoles on a reduced size ground plane is explored by Richard [37].

A standard antenna has gain-pattern and input-impedance characteristics that are known with a small degree of uncertainty. Standard antennas are typically used in gain-transfer measurements to determine the gain of test antennas by comparing the signal level received by the test antenna to that received by the standard antenna when it replaces the test antenna on the antenna test range.

For simplicity and accuracy, the standard antenna should have polarization and gain patterns similar to those of the test antenna. The primary contribution of the measurement effort described in this paper is applying Altshuler's resistive loading technique [22] to make a small ground plane appear to be electrically larger than it actually is. An infinite perfectly conducting ground plane is assumed for the calculations. The difference between measured and calculated data gives a reasonable estimate of measurement uncertainty using a finite ground plane.

The appropriate form for the surface impedance of a radial-wire ground screen is utilized to obtain quantitative effects of the important parameters (number of radials, length of radials, wire radius, and ground constants) on the low-angle radiation of a vertical antenna [38]. Wire ground screens are commonly used to enhance low-angle radiation from HF antennas.

In this paper, researchers considered a radial-wire ground screen which has a variable impedance as a result of its variable wire spacing. Through the use of such a ground screen, the dependence of the low-angle radiation on the important parameters (number, length, and radius of radials) can be directly examined. It was found that the effect of currents reflected

from the edge of the screen is quite small when the screen is reasonably large. However, the situation is not quite so clear-cut for smaller screens, and a numerical integration is required to yield quantitative results.

2.4. Source Models and Implications

The first step in the analysis of the fields radiated by an antenna is the specification of the sources to excite the antenna. The delta gap model, magnetic frill generator and plane wave incidence are the main source modelings for thin wire antennas [10]. The delta-gap source modeling is the simplest and most widely used of the two, but it is also the least accurate, especially for impedances. Usually it is most accurate for smaller width gaps [18].

The thin wire approximate kernel results from taking an axis to surface distance. This is because of the difficulty in establishing the far field radiation on the antenna surface. Although with the approximate Kernel, reliable results have been produced so long as the thin wire criteria are not exceeded. The effect of exceeding the thin wire criteria leaves the graph of the current distributions with oscillations. These oscillations are actually due to round off errors which results when large segmentation is applied on the linear conductor. It is also seen in this work that the thin wire approximation deteriorate as the segment length approaches the radius of the wire. These oscillations are not peculiar to Pocklington's equation alone. It is also applicable to the Hallen's equation counterpart [12].

2.5. Wires of Arbitrary shapes

An Important engineering problem is the electromagnetic behaviour of thin wires objects. The distinction between antennas and scatterers is primarily the location of the source. When the source is on the object it is viewed as an antenna; if the source is distant from the object it is viewed as a scatterer. Therefore the development of the solution may be easily followed. Basically it involves an approximation of the exact equation for conducting bodies by an approximate equation valid for thin wires. A replacement of the derivatives by finite difference approximations, yielding an approximate operator. The use of pulse functions for expansion functions, to give a step approximation to the current and charge yields satisfactory result.

To effect a solution, the wire is considered as N- short segment connected together. The end points of each segment define a pair of terminals in space. These N pairs of terminals can be thought of as forming an N-port network, and the wire object is obtained by short-circuiting all ports of the network [39]. One can determine the impedance matrix for the N-port network by

applying a current source to each point in turn, and calculating the open circuit voltage at all the ports. This procedure involves only current elements in empty space. The admittance matrix is the inverse of the impedance matrix. Once the admittance matrix is known, the port currents (current distribution on the wire) are found for any particular voltage excitation (applied field) by matrix multiplication [39].

The problem of finding the current distributions on the wire antenna or scatterer elements is but a particular case of the general boundary value problem involving conducting bodies in a known impressed field, $\bar{E}_i$. The boundary condition at the surface of each perfect conductor is

$$\hat{n} \times \bar{E}^t = 0 \tag{2.4}$$

where $\hat{n}$ is a unit vector normal to the surface of the conductor and in the outward direction, and $\bar{E}_t$ is the total electric field vector consisting of both incident and scattered fields. The scattered field $\bar{E}_s$ is defined as the field produced by all currents and charges on the conductors The equations that summarize this boundary value problem are given below [40]:

$$\mathbf{E} = -\nabla\phi_e - j\omega\boldsymbol{A} \tag{2.5}$$

$$\boldsymbol{A} = \frac{\mu}{4\pi}\oiint_s \boldsymbol{J}\frac{e^{-jkR}}{R}\boldsymbol{ds}. \tag{2.6}$$

$$\boldsymbol{\Phi} = \frac{1}{4\pi\varepsilon}\oiint_s \boldsymbol{\sigma}\frac{e^{-jkR}}{R}\boldsymbol{ds} \tag{2.7}$$

$$\boldsymbol{\sigma} = \frac{\mathbf{1}}{\boldsymbol{j\omega}}\nabla.\boldsymbol{J} \tag{2.8}$$

Computer programs are presented for analysis of the electromagnetic behaviour of arbitrarily bent, thin wire antennas and scatterers [8].

For radiation problems, the current distributions on the wires are determined along with appropriate near and far field patterns and input admittances corresponding to the feed points. For plane wave scattering problems, the current distributions are again computed along with appropriate scattered fields and bistatic radar cross- section patterns.

2.6. Method of Moments Analysis

The method of moments technique, as applied to the problems in electromagnetic theory, was introduced by Harrington. [30]. A unified treatment of matrix methods useful for field problems is given. The basic mathematical concept is the method of moments, by which the functional equations of field theory are reduced to matrix equations.

Consider the generalized problem defined as [41] [42]:

$$L(f) = g \tag{2.9}$$

where L is a linear operator, g a known forcing function, and f an unknown.

L is typically an integro-differential operator, f an unknown function (current) and g a known driving function (incident field).

Expanding f into a sum of N basis functions yields:

$$f = \sum_{n=1}^{N} a_n f_n \tag{2.10}$$

where a_n are unknown coefficients. Owing to the fact that L is linear, and substituting (2) into (1) yields:

$$\sum_{n=1}^{N} a_n L(f_n) \approx g \tag{2.11}$$

where the residue is:

$$R = g - \sum_{n=1}^{N} a_n L(f_n) \tag{2.12}$$

In (3), the basis functions are usually chosen in a way that the expected behaviour of the unknown function is modeled throughout its domain. Depending on the problem, the basis functions may be scalar or vector, and if the basis functions exists in the domain, they are called local or sub-sectional basis functions. However, if the support spans the entire problem domain, then they are called global or entire-domain basis functions.

If we define an inner product or moment between a basis function $f_n\left(\mathbf{r}'\right)$ and a testing function $f_m\left(\mathbf{r}\right)$ [31],[32].

$$\left\langle f_m, f_n \right\rangle = \int_{f_m} f_m\left(\mathbf{r}\right) \cdot \int_{f_n} f_n\left(\mathbf{r}'\right) d\mathbf{r}' d\mathbf{r} \tag{2.13}$$

where the integrals could be line, volume, or surface integrals depending on the basis and testing functions. If we impose an additional condition that the product of each testing function with the residual function be zero, we obtain:

$$\sum_{n=1}^{N} a_n \left\langle f_m, L\left(f_n\right)\right\rangle = \left\langle f_m, g \right\rangle \tag{2.14}$$

Which transforms to $N \times N$ matrix equation [Z][A] = [B] with matrix elements:

$$Z_{mn} = \left\langle f_m, L\left(f_n\right)\right\rangle \tag{2.15}$$

with right-hand side vector elements as:

$$b_m = \left\langle f_m, g \right\rangle \tag{2.16}$$

For many applications, the choice of the testing function is critical in obtaining good solutions, and one of the most commonly used is the method of Galerkin in which same basis functions are used as the testing functions. The advantage of this approach is that boundary condition is enforced throughout the solution domain instead at discrete points as with point matching. This

solution may be exact or approximate, depending upon the choice of the f_n and w_n. The particular choice $w_n = f_n$ is known as Galerkin's method [3].

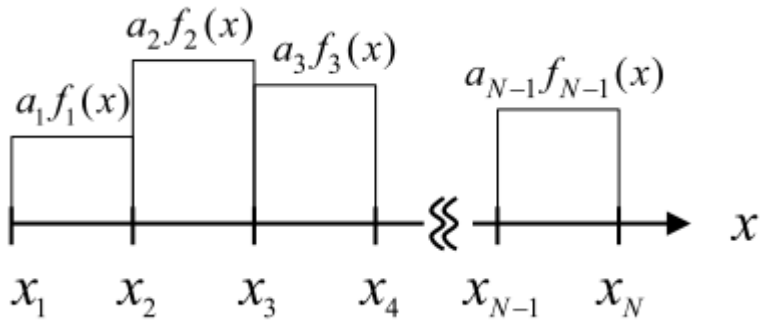
P. Banjerjee and T. Bezboruah [18], theoretically investigate current distribution, radiation patterns, directivity, radiation resistance, input impedance, of finite-length dipole antenna using simulation software Matlab [18]. They propose to simulate and study the parameters of finite length dipole, such as current distribution, radiation pattern, radiation resistance, and directivity in an unbounded medium.

However, they used piecewise constant sub-domain functions and point - matching. They have used Pocklington's integro - differential equation and Hallen's integral equation to find the current distribution on conducting wires, and then the values of input impedance obtained from two methods are compared. In all the cases, theoretical results are found to be closely in agreement with the experimental results..

Matrix methods are applied for analysis and design of long thin- wire antennas with multiple excitations [43]. The purposes of this communication are to show how matrix methods can be applied to problems involving a single wire with multiple excitations, to point out a simple procedure for including antenna loading in the computations and to show that in some simple cases, passive loading can be used instead of independent excitations to achieve some desired radiation or scattering characteristics. The computer programs suitable for handling the procedures described are available [39]. The Electromagnetic (EM) modeling and design of antenna systems are been the subject of extensive research in the last three decades [42]. This paper represents the method of moments for the evaluation of the charge distribution and capacitance of conducting structures used in high frequency integrated circuits. The square shape of sub-domains was chosen because of its ability to conform easily to any geometrical surface or shape and at the same time to maintain simplicity of approach compared to the another shaped modeling. A computer programs based on the formulations was developed to determine the charge distributions, hence capacitance of the conducting structures. This method can be used for the determination of equivalent circuit models of multiconductor or multiwire arrangement used in electronics systems.

2.6.1. Types of Basis Functions used in Method of Moments (MoM) Problems

Pulse Functions: this is a set of pulse basis functions as shown below, where the antenna wire function is divided into N points comprising of $N-1$ sub-segments. In the depiction below, all the segments are assumed to be of equal lengths [40].



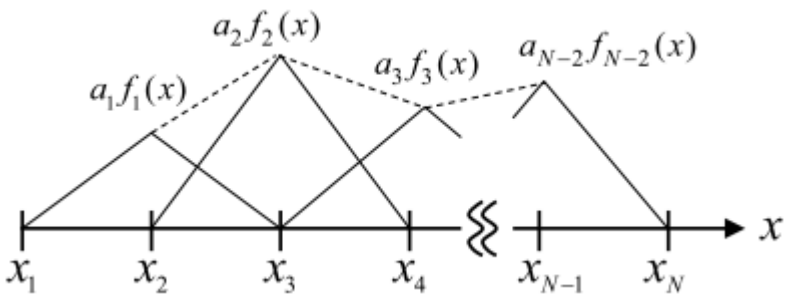
Mathematically, the pulse function is defined as [40] [44] [45]:

$$f_n(x) = \begin{cases} 1, x_n \le x \le x_{n+1} \\ 0, elsewhere \end{cases} \tag{2.17}$$

In each segment, the pulse function provides a simple and a crude approximation to the solution; however, they greatly simplify the evaluation of MoM matrix elements, but they cannot be used when the integral operator contains a derivative because it is known in literature that the derivative of a pulse function is impulsive.

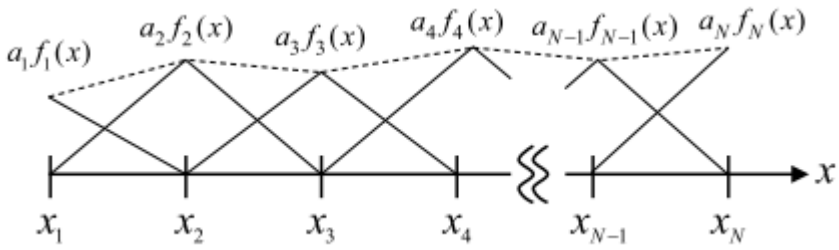
Piecewise Triangular Functions

Unlike pulse functions, a triangular function as depicted below spans two segments and shows variation of zero at the outer points to unity at the center. The domain is divided into N points and $N-1$ sub-segments, resulting in $N-2$ basis functions.



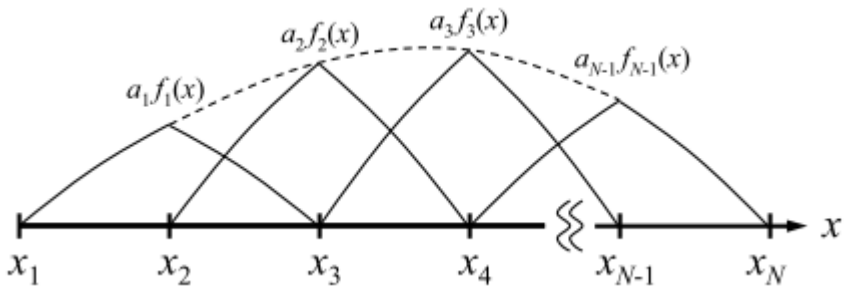
Owing to the fact that adjacent functions do overlap by one segment, triangular functions always provide a piecewise variation of a solution between segments.

One major advantage of piecewise triangular function is that it can be applied when the integral operator contains a derivative with respect to x; this is very important in the redistribution of vector derivatives. It should be noted that the function shown above assumes that the value of the solution at the ends of the domain is zero. When this is not the case, half triangles are assigned to the first and last segments thereby removing the condition that solution must be zero. This is depicted below [44]:



Piecewise Sinusoidal Functions

The piecewise sinusoidal functions are used in the analysis of wire antennas due to their ability to represent current distributions that are sinusoidal. The figure below shows this class of functions:



Within the method of ***moments procedure*** a linear current approximation is used together with the ***method of subsections***.

Hence, triangle current expansion functions are used where each function is non-zero only over a relatively small portion of a wire. Triangle functions are also used for testing functions

(Galerkin's procedure). These choices were made because of the observation of Harrignton and Mautz [48] that the solution converges about twice as fast as the solution resulting from use of pulses for current expansions and point matching.

CHAPTER THREE

METHODOLOGY

3.1. Introduction

This project analyses the electromagnetic behaviour of monopole antennas in which the following quantities are determined for a monopole antenna:

1. The current distributions
2. Radiation patterns

An antenna can be a combination of two or more thin wires with different shapes and radii. These thin wires can be loaded and/or excited at any points along their lengths. The following assumptions are made in this work [8]

i. The radius (r) of the thin wire antenna must be smaller than the length (l) of the antenna such that $r << l$.
ii. A minimum of twenty segments should be used per wavelength.
iii. Each segment should be at least five times the radius of the wire (5r).

3.2. Method of Moments

Method of Moments (MoM) is a technique for estimating parameters. Its popularity is due to its computational simplicity [28] . Let us consider an inhomogenous linear equation.

$$L(f) = g \tag{3.1}$$

where L is a linear operator and g is a function with known values while f is a function to be determined. Let f be the summation of $\alpha_1 f_1, \alpha_2 f_2, \alpha_3 f_3, \dots, \alpha_n f_n$ in the L's domain. Mathematically, this can be expressed as [28]

$$f = \sum_n \alpha_n f_n \tag{3.2}$$

where α_n are constants and f_n are referred to as the expansion (or basis) functions. The n in equation (3.2) is a finite number for approximate solutions while it is infinite for exact solutions. Generally, the higher the value of n, the higher the accuracy of the solution obtained. Substituting the value of f in equation (3.2) into equation (3.1), we have

$$\sum_n \alpha_n L(f_n) = g \tag{3.3}$$

Taking the inner product of equation (3.3), we have [28]

$$\sum_n \alpha_n \langle w_m, Lf_n \rangle = \langle w_m, g \rangle \tag{3.4}$$

where w_m are the weighting (or testing) functions.

Expressing equation (3.4) in the matrices format, we have

$$[l_{mn}][\alpha_n] = [g_m] \tag{3.5}$$

where $[l_{mn}]$ is a $m \times n$ matrix, $[\alpha_n]$ is a n-column vector and $[g_m]$ is a m-column vector expressed as:

$$[l_{mn}] = \begin{bmatrix} \langle w_1, Lf_1 \rangle & \langle w_1, Lf_2 \rangle & \cdots & \langle w_1, Lf_n \rangle \\ \langle w_2, Lf_1 \rangle & \langle w_2, Lf_2 \rangle & \cdots & \langle w_2, Lf_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle w_m, Lf_1 \rangle & \langle w_m, Lf_2 \rangle & \cdots & \langle w_m, Lf_n \rangle \end{bmatrix} \tag{3.6}$$

$$[\alpha_n] = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} \tag{3.7}$$

$$[g_m] = \begin{bmatrix} \langle w_1, g \rangle \\ \langle w_2, g \rangle \\ \vdots \\ \langle w_m, g \rangle \end{bmatrix} \tag{3.8}$$

The n-column vector $[\alpha_n]$ can be determined from equation (3.5) as

$$[\alpha_n] = [l_{mn}]^{-1}[g_m] \tag{3.9}$$

If matrix $[l_{mn}]$ is nonsingular, then its inverse exists. Equation (3.2) can be expressed in matrix form as

$$f = [f_n][\alpha_n] = [f_n][l_{mn}]^{-1}[g_m] \tag{3.10}$$

where $[f_n]$ is a n-row vector expressed as

$$[f_n] = [f_1 \quad f_2 \quad \cdots \quad f_n] \tag{3.11}$$

The choice of expansion functions, f_n, and weighting functions, w_m, determines whether the solution will be exact or approximate. When a choice of $w_m = f_n$ is made, then it is referred to as Galerkin's method. The accuracy of the solution desired, the size of the matrix $[l_{mn}]$ that can be inverted and the realization of well-conditioned matrix $[l_{mn}]$ will determine the choice of f_n and w_m. Generally,

i. f_n should be chosen such that it is linearly independent and the superposition of $f = \sum_n \alpha_n f_n$ can reasonably estimate f.

ii. w_m should be linearly independent such that the inner products $\langle w_m, g \rangle$ depend on relatively independent properties of g.

3.3. Problem Formulation

Electromagnetic (EM) waves travel at approximately constant speed of 3 x 10^6 m/s in free space. This speed is referred to as the speed of light (c). The important equation that relates the frequency of oscillation and its wavelength is given in equation (3.12)

$$c = f\lambda \tag{3.12}$$

where f is the frequency of oscillation and λ is the corresponding wavelength. The propagation constant of the EM wave is given in equation (3.13)

$$\beta = {}^{2\pi}/_{\lambda} \tag{3.13}$$

Where β is the propagation constant.

The current distribution on a thin-wire antenna is a special case of the general boundary value problem. The starting point is the four Maxwell's equations from (3.14) through (3.17).

Please find the proof in the appendix.

$$\boldsymbol{\nabla} \cdot \boldsymbol{D} = \rho_v \tag{3.14}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \tag{3.15}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{3.16}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{J} \tag{3.17}$$

Because $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$, then the Maxwell's second equation, follows that

$$\frac{A}{B} = \boldsymbol{\nabla} \times \boldsymbol{A} \tag{3.18}$$

where **A** is a vector magnetic potential. Substituting equation (3.17) into equation (3.15), then $\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial(\nabla \times \boldsymbol{A})}{\partial t} = -\boldsymbol{\nabla} \times \frac{\partial \boldsymbol{A}}{\partial t}$. Therefore,

$$\boldsymbol{\nabla} \times \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t}\right) = 0 \tag{3.19}$$

Replacing the curl of equation (3.18) with the gradient of a scalar quantity, we have $\boldsymbol{\nabla}\psi + \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t}\right) = 0$. Therefore, $\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} = -\boldsymbol{\nabla}\psi$, where ψ is the scalar electric potential [10].

This implies that $\boldsymbol{E} = -\left(\frac{\partial \boldsymbol{A}}{\partial t} + \boldsymbol{\nabla}\psi\right)$. When a time-harmonic variation of the form, $\frac{\partial}{\partial t} = j^{wt}$. Therefore, the scattered field $\boldsymbol{E^s}$, produced by the charge density $\boldsymbol{\sigma}$ and the current density $\boldsymbol{J}$, is expressed in terms of retarded vector $\boldsymbol{A}$ and scalar ψ potential integrals:

$$\boldsymbol{E^s} = -(\boldsymbol{\nabla}\psi + jw\boldsymbol{A}) \tag{3.20}$$

In a far-field pattern, $\boldsymbol{\nabla}\psi = 0$, hence, $\boldsymbol{E^s} = -jw\boldsymbol{A}$. The vector potential **A** for an electric current density **J** can be expressed as shown in equation (3.21).

$$\boldsymbol{A} = \frac{\mu}{4\pi}\oiint \boldsymbol{J}\frac{e^{-jkR}}{R}ds \tag{3.21}$$

Where μ is the permeability and R is the distance from the source point to the field point. Similarly, the scalar potential ψ for an electric charge density $\boldsymbol{\sigma}$ can be expressed as:

$$\psi = \frac{1}{4\pi\varepsilon}\oiint \sigma\frac{e^{-jkR}}{R}ds \tag{3.22}$$

$\int \frac{e^{-jkR}}{R}$ is referred to as the Green's function. The relationship between the charge density and the current density can be expressed as [10]:

$$\sigma = -\frac{1}{j\omega}\boldsymbol{\nabla}.\boldsymbol{J} \tag{3.23}$$

And the imposition of the boundary condition $n \times (\boldsymbol{E^i} + \boldsymbol{E^s}) = 0$ yields:

$$n \times \boldsymbol{E^s} = -n \times \boldsymbol{E^i} \tag{3.24}$$

where $\boldsymbol{E^i}$ is the impressed field.

In the case of thin wires, the current density $\boldsymbol{J}$ can be approximated by filaments of current $\boldsymbol{I}$ and charge density can be approximated by charge $\boldsymbol{\sigma}$ on the wire axis and assuming the current flows only in the axial direction, equations (3.19) through (3.23) become [10]:

$$-\boldsymbol{E}_l^i = -jw\boldsymbol{A} - \frac{\partial\psi}{\partial l} \tag{3.25}$$

$$\boldsymbol{A} = \frac{\mu}{4\pi}\int \boldsymbol{I}(l)\frac{e^{-jkR}}{R}dl \tag{3.26}$$

$$\psi = \frac{1}{4\pi\varepsilon}\int \sigma(l)\frac{e^{-jkR}}{R}dl \tag{3.27}$$

$$\sigma = -\frac{1}{j\omega}\frac{dI}{dl} \tag{3.28}$$

E^i is the impressed field.

3.4. Current Distribution

The generalized voltage matrix [V] of dimension $M \times 1$ can be expressed as

$$[V] = [Z][I] \tag{3.29}$$

where $[Z]$ is the generalized impedance matrix of dimension $M \times M$, $[I]$ is the generalized current matrix of $M \times 1$. M is the number of expansion functions. Equation (3.29) is analogous to equation (3.1) where $[Z]$ is the operator, $[V]$ is a given function and $[I]$ is a function to be determined. Therefore, the desired solution

$$[I_n] = [Z_{mn}]^{-1}[V_m] \tag{3.30}$$

Comparing equation (3.30) to equation (3.9), $[I_n]$ is analogous to $[\alpha_n]$, $[Z_{mn}]$ is analogous to $[l_{mn}]$ and $[V_m]$ is analogous to $[g_m]$. m and n are segments of thin antenna wire with N total segments. Generally, admittance Y and impedance Z are inversely proportional. That is, $[Y] = [Z]^{-1}$. Therefore, $[I] = [Y][V]$. $[Y]$ is the generalized admittance matrix which is independent of excitations but a function of the geometry of the problem and operating frequency.

In this project, the subscript n, is replaced with i, n while the subscript m is replaced with j, m. This implies that $[I_n]$ becomes $[I_{i,n}]$, $[Z_{mn}]$ becomes $[Z_{j,m,i,n}]$ and $[V_m]$ becomes $[V_{j,m}]$. where $i = 1,2, \cdots NW$, $j = 1,2, \cdots NW$, $n = 1,2, \cdots NE(i)$ and $m = 1,2, \cdots NE(j)$. NW is the number of wires that makes up the thin-wire antenna, $NE(i)$ is the number of expansion functions of the ith wire.

Similar to equation (3.2) $f = \sum_n \alpha_n f_n$,

$$I = \sum_i \sum_n I_{i,n} f_{i,n} \tag{3.31}$$

Substituting equation (31) into (29), we have

$$\sum_i \sum_n I_{i,n} Z f_{i,n} = V \tag{3.32}$$

Taking the inner product of equation (3.32), we have

$$\sum_i \sum_n I_{i,n} \langle w_{j,m} Z f_{i,n} \rangle = \langle w_{j,m}, V \rangle \tag{3.33}$$

Expressing equation (3.33) in matrix form, we have

$$[Z_{j,m,i,n}][I_{i,n}] = [V_{j,m}] \tag{3.34}$$

where

$$[Z_{j,m,i,n}] = \begin{bmatrix} \langle w_{1,1}, Zf_{1,1}\rangle & \cdots & \langle w_{1,1}, Zf_{1,NE(1)}\rangle & \langle w_{1,1}, Zf_{2,1}\rangle & \cdots & \langle w_{1,1}, Zf_{NW,NE(NW)}\rangle \\ \langle w_{1,2}, Zf_{1,1}\rangle & \cdots & \langle w_{1,2}, Zf_{1,NE(1)}\rangle & \langle w_{1,2}, Zf_{2,1}\rangle & \cdots & \langle w_{1,2}, Zf_{NW,NE(NW)}\rangle \\ \vdots & \cdots & \vdots & \vdots & & \cdots \ \vdots \\ \langle w_{1,NE(1)}, Zf_{1,1}\rangle & \cdots & \langle w_{1,NE(1)}, Zf_{1,NE(1)}\rangle & \langle w_{1,NE(1)}, Zf_{2,1}\rangle & \cdots & \langle w_{1,NE(1)}, Zf_{NW,NE(NW)}\rangle \\ \langle w_{2,1}, Zf_{1,1}\rangle & \cdots & \langle w_{2,1}, Zf_{1,NE(1)}\rangle & \langle w_{2,1}, Zf_{2,1}\rangle & \cdots & \langle w_{2,1}, Zf_{NW,NE(NW)}\rangle \\ \vdots & \cdots & \vdots & \vdots & & \cdots \ \vdots \\ \langle w_{NW,NE(NW)}, Zf_{1,1}\rangle & \cdots & \langle w_{NW,NE(NW)}, Zf_{1,NE(1)}\rangle & \langle w_{NW,NE(NW)}, Zf_{2,1}\rangle & \cdots & \langle w_{NW,NE(NW)}, Zf_{NW,NE(NW)}\rangle \end{bmatrix} \tag{35}$$

$$[I_{i,n}] = \begin{bmatrix} I_{1,1} \\ I_{1,2} \\ \vdots \\ I_{1,NE(1)} \\ I_{2,1} \\ \vdots \\ I_{NW,NE(NW)} \end{bmatrix} \tag{3.36}$$

$$[V_{j,m}] = \begin{bmatrix} \langle w_{1,1}, V\rangle \\ \langle w_{1,2}, V\rangle \\ \vdots \\ \langle w_{1,NE(1)}, V\rangle \\ \langle w_{2,1}, V\rangle \\ \vdots \\ \langle w_{NW,NE(NW)}, V\rangle \end{bmatrix} \tag{3.37}$$

3.5. Determination of the Generalized Impedance Matrix

Assuming an antenna wire is divided into N segments such that the two ends are represented as $\overset{-}{N}$ and $\overset{+}{N}$ as shown in Figure 3-1. The starting point of Nth segment is $\overset{-}{N}$ while the end point is $\overset{+}{N}$. The midpoint is labelled N [28].

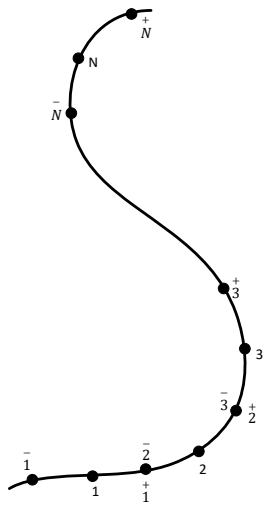


Figure 3-1: Antenna wire with N segments

Let us further assume that we have an antenna wire such that $i = j$, as shown in Figure 3-2. The nth wire has the starting and ending points labelled as $\overset{-}{n}$ and $\overset{+}{n}$ while the starting and ending points of the mth antenna wire segment are $\overset{-}{m}$ and $\overset{+}{m}$. The midpoints of the nth and mth segments are n and m. Therefore, the length of the nth wire segment is Δl_n which measures the distance between $\overset{+}{n}$ and $\overset{-}{n}$. Moving half-distance after the nth wire segment gives $\Delta l_{\overset{+}{n}}$ while moving half-distance before the nth wire segment gives $\Delta l_{\overset{-}{n}}$. This implies that $\Delta l_{\overset{+}{n}}$ measures the distance between $(n+1)$ and n and $\Delta l_{\overset{-}{n}}$ measures the distance between $(n-1)$ and n [28].

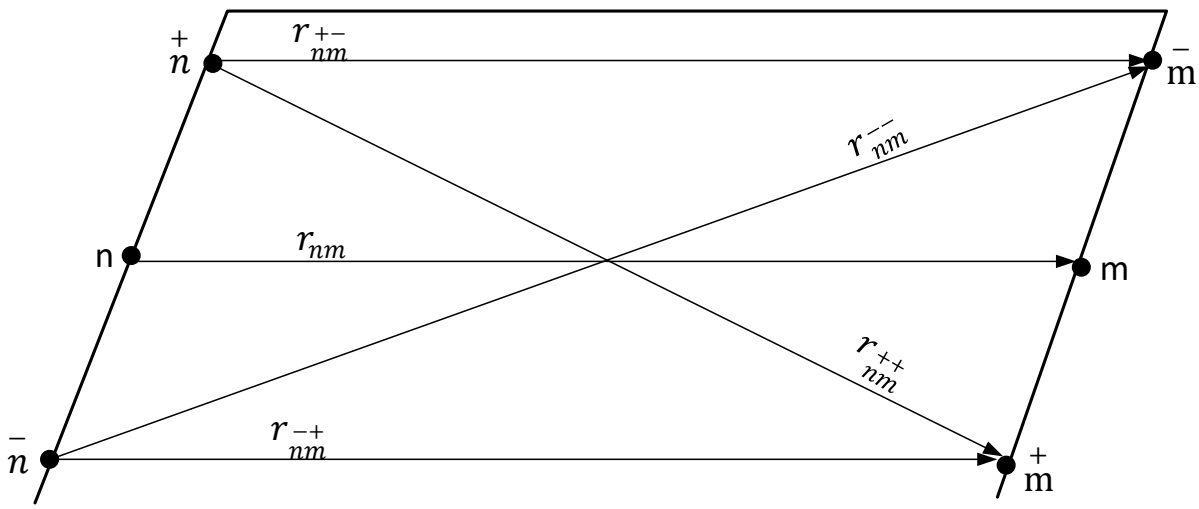


Figure 3-2: nth and mth segments of N segments of an antenna wire

Following from equations (3.25) to (3.28), we have

$$-\boldsymbol{E}_l^i(m) = -jw\boldsymbol{A}(m) - \frac{\psi(\bar{m}) - \psi\left(\overset{+}{m}\right)}{\Delta l_m} \tag{3.38}$$

$$\boldsymbol{A}(m) = \frac{\mu}{4\pi}\sum_n I(n) \int_{\Delta l_n} \frac{e^{-jkR}}{R} dl \tag{3.39}$$

$$\psi\left(\overset{+}{m}\right) = \frac{1}{4\pi\varepsilon}\sum_n \sigma\left(\overset{+}{n}\right) \int_{\Delta l_{\overset{+}{n}}} \frac{e^{-jkR}}{R} dl \tag{3.40}$$

$$\sigma\left(\overset{+}{n}\right) = -\frac{1}{j\omega}\frac{I(n+1) - I(n)}{\Delta l_{\overset{+}{n}}} \tag{3.41}$$

Equations (3.39) and (3.40) have similar integrals. Therefore, let the similar expressions be represented with φ such that [28]:

$$\varphi(n,m) = \frac{1}{4\pi\Delta l_n}\sum_n \int_{\Delta l_{\overset{+}{n}}} \frac{e^{-jkR}}{R} dl \tag{3.42}$$

The charge density $\sigma = {}^{q}/_{\Delta l}$, therefore, $q = \sigma\Delta$, where q is the charge filament. It follows then that $q\left(\overset{+}{n}\right) = \frac{1}{jw}I(n)$ and $q(\bar{n}) = -\frac{1}{jw}I(n)$.

Hence, equation (39) which is the vector potential at m due to $I(n)$ can be expressed as

$$\boldsymbol{A}(m) = \mu I(n)\Delta l_n\, \varphi(n,m) \tag{3.43}$$

The scalar potential at $\overset{+}{m}$ can be expressed as

$$\psi\left(\overset{+}{m}\right) = \frac{1}{\varepsilon}\,\sigma\left(\overset{+}{n}\right)\Delta l_n\left[\varphi\left(\overset{+}{n},\overset{+}{m}\right) - \varphi\left(\bar{n},\overset{+}{m}\right)\right] \tag{3.44}$$

Substituting $\sigma\left(\overset{+}{n}\right) = \frac{1}{jw\Delta l_n}I(n)$ into equation (44), we have

$$\psi\left(\overset{+}{m}\right) = \frac{1}{jw\varepsilon}\left[I(n)\varphi\left(\overset{+}{n},\overset{+}{m}\right) - I(n)\varphi\left(\bar{n},\overset{+}{m}\right)\right] \tag{3.45}$$

Similarly, the scalar potential at $\bar{m}$ can be expressed as

$$\psi(\bar{m}) = \frac{1}{jw\varepsilon}\left[I(n)\varphi\left(\overset{+}{n},\bar{m}\right) - I(n)\varphi(\bar{n},\bar{m})\right] \tag{3.46}$$

Recall that $V(m) = E^i(m)\Delta l_m$ and therefore, $Z_{mn} = {V(m)}/{I(n)}$. Substituting equations (3.43), (3.45) and (3.46) into equation (3.38), we have

$$\begin{aligned} \boldsymbol{E}^i(m)\Delta l_m = {} & jw\mu I(n)\Delta l_n \Delta l_m\, \varphi(n,m) \\ & - \frac{1}{jw\varepsilon} I(n)\left[\varphi\left(\overset{+}{n},\overset{-}{m}\right) - \varphi\left(\overset{-}{n},\overset{-}{m}\right)\right] \\ & + \frac{1}{jw\varepsilon} I(n)\left[\varphi\left(\overset{+}{n},\overset{+}{m}\right) - \varphi\left(\overset{-}{n},\overset{+}{m}\right)\right] \end{aligned} \tag{3.47}$$

Therefore [28],

$$\begin{aligned} Z_{mn} = {} & jw\mu \Delta l_n \Delta l_m\, \varphi(n,m) \\ & + \frac{1}{jw\varepsilon}\left[\varphi\left(\overset{+}{n},\overset{+}{m}\right) - \varphi\left(\overset{-}{n},\overset{+}{m}\right) - \varphi\left(\overset{+}{n},\overset{-}{m}\right) + \varphi\left(\overset{-}{n},\overset{-}{m}\right)\right] \end{aligned} \tag{3.48}$$

In this project, it is assumed that more than one wire can make up an antenna where i and j represent the number of wires. Then Z_{mn} is replaced with Z_{jmin}. In equation (3.48), the only unknown is $\varphi(n,m)$ which can be evaluated as given in Asianuba and Nzeako [9].

The realization of a well-conditioned impedance matrix $[Z]$ necessitates the use of Galerkin's method with triangle current expansion functions, with piecewise linear expansion, as shown in Figure 3-3.

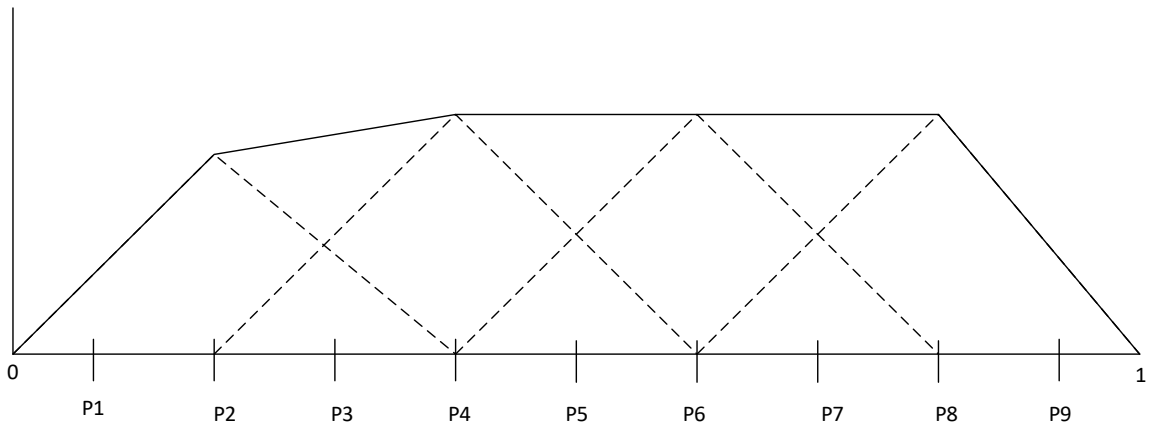


Figure 3-3: Triangle expansion functions with piecewise linear approximation

It can be seen that each triangle function $(\boldsymbol{T}_{j,m})$ is non-zero over four consecutive segments. where a segment $l_{i,j}$ is the distance between two points, say $P_{i,j}$ and $P_{i,j+1}$ in Figure 3-4. where $P_{i,j}$ is the point on the ith wire for the jth point on the wire for the total number of points, NP. Therefore, the total number of segments, $NS = NP - 1$ [8].

Successive triangles overlap the previous two segments except at the ends of the wire. Therefore, currents at the ends of the wires equal zero. The total number of expansion functions is given as

$$NE = \frac{NP - 3NW}{2} \tag{3.49}$$

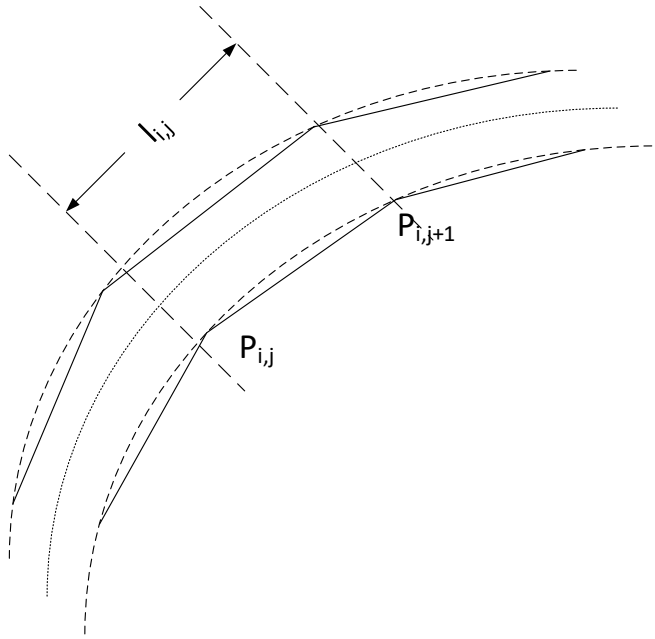


Figure 3-4: Approximation of a bent, thin wire

3.6. Radiation Patterns

Towards obtaining the radiation pattern of a wire antenna, the antenna is treated as an array of N current elements $I(n)\Delta l_n$. Analogous to equation (3.39), the far-zone vector potential can be expressed as [8]

$$\boldsymbol{A} = \frac{\mu e^{-jkr_0}}{4\pi r_0} \sum_n I(n)\, \Delta l_n e^{jkr_n \cos \xi_n} \tag{3.50}$$

where r_0 is the radius vector to the distant field point, r_n is the radius vector to the source point and ξ_n are the angles between r_0 and r_n. The far-zone field components are similar to equation (3.25) such the scalar potential part of the equation is zero. That is, $\frac{d\psi}{dl} = 0$. Therefore,

$$\boldsymbol{E}_{\theta} = -jw\boldsymbol{A}_{\boldsymbol{\theta}};\ \boldsymbol{E}_{\phi} = -jw\boldsymbol{A}_{\boldsymbol{\phi}} \tag{3.51}$$

where θ are the conventional coordinate angles and ϕ are the spherical coordinate angles.

3.7. Implementation Tool

MATLAB version 9.8 (R2020a) was used in implementing this algorithm. Although FORTRAN can also perform matrix operations pretty well, MATLAB was specially built for matrices operation. Fewer lines of codes are also needed when compared with similar implementation in FORTRAN. Since this is a research project, MATLAB performs excellently well in implementing the Method of Moments where parameters are represented in matrices form. Furthermore, MATLAB contains many inbuilt functions and toolboxes which proved to be helpful.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

4.1. Introduction

This section of the project dissertation discusses the results obtained for three different scenarios: reactively loaded, resistively loaded and combination of resistive and reactive loads.

The loads can be applied at any point along the length of the monopole antenna. The thin-wire monopole antenna comprises of three wires of different radii. The radii of the wires are 0.00337 m (R_1), 0.0042 m (R_2) and 0.0052 m (R_3).

It is assumed that R_2 is 125% of R_1 and R_3 is 125% of R_2. The straight thin-wire monopole antenna is z-directed with its centre along the z-axis origin. It is 0.9 m long. The wires with radii R_2 and R_3 are each 0.291 m long while the wire with radius R_1 is 0.318 m long. The largest radius (0.0052 m) is much less than the length of the wire antenna (0.9 m). Each segment is 0.0265 m which is greater than five times the largest radius (0.0260 m). The wavelength varies from 1 m to 2 m which is greater than 20 times the segment (0.53 m)

The straight monpole wire antenna can have single or multiple excitations along its length.

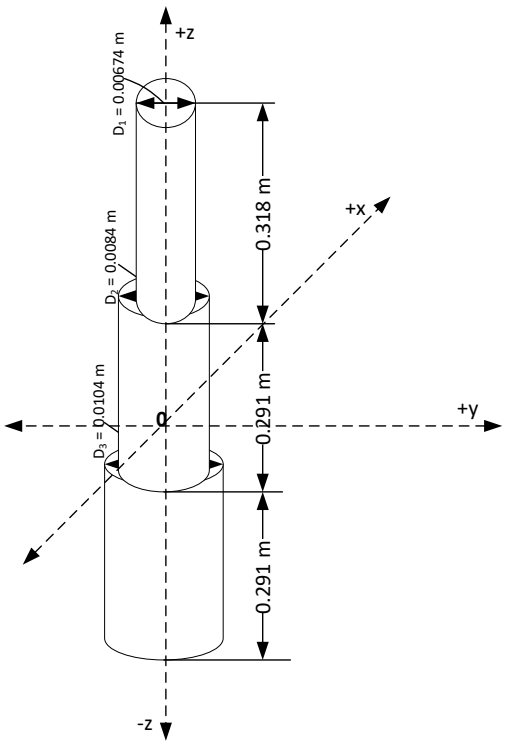


Figure 4-1: The z-directed monopole antenna dimensions

4.2. Scenario 1 – Reactively Loaded

The monopole wire antenna is singly excited by 1V feed voltage at 0.4365 m of the antenna length and loaded reactively at 0.45m with j100Ω.

Figure 4-2 shows the current distribution for 300 MHz, 320 MHz, 340 MHz and 360 MHz frequencies for the single excitation. It can be seen that the magnitude of the current distribution increases with increasing frequency. Using triangular expansion function in equation (3.49), 0.9 m monopole antenna has 13 triangles with the voltage fed at the apex of the 7th triangle coinciding with 0.45 m of the antenna [8]. The peak of the current distribution is the point of excitation and it decreases rapidly to either side of the length of the antenna.

The current distribution values on either sides of the point of excitation are not completely symmetrical.

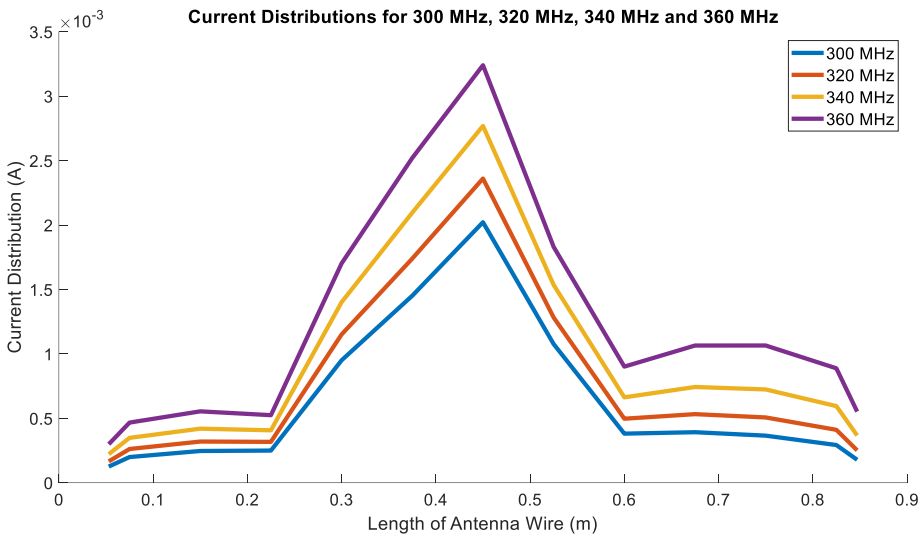


Figure 4-2: Current distributions for 300, 320, 340 and 360 MHz frequencies single excitation

Figure 4-3 shows that as the frequency increases, more propagating distances are covered but the angular coverage decreases. It represents a plot of the magnitude of the electric field as a function of the anglar space. For example, while the propagating distance of 300 MHz frequency is 0.08, 360 MHz frequency propagated to a distance of 0.2 but the angular coverage of 360 MHz frequency is less than that of 300 MHz represented by a narrower field pattern. The propagating distance is a measure of the directivity of the antenna.

The directivity of an antenna in a given direction is defined as "the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all

directions. The average radiation intensity is equal to the total power radiated by the antenna divided by 4π" [49]. It is a dimensionless antenna parameter.

Therefore, it can be stated that the monopole antenna with a unit voltage at almost the centre of the antenna has higher directivity (more intense radiation) at $\theta = \pi/2$ for the frequencies 300 MHz, 320 MHz, 340 MHz and 360 MHz

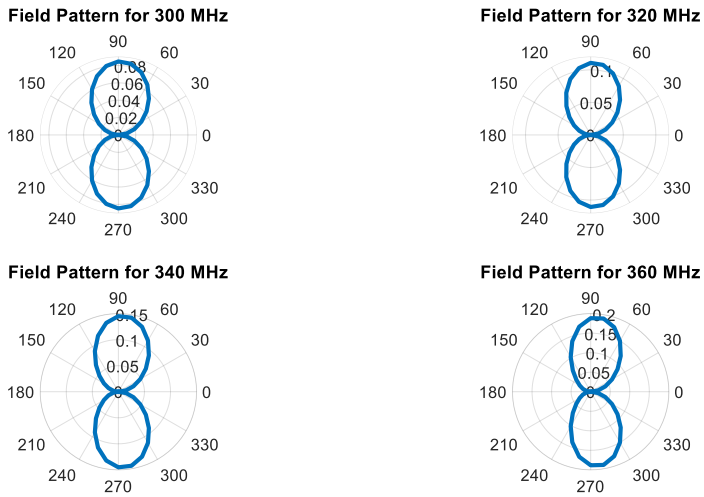


Figure 4-3: Field patterns for 300, 320, 340 and 360 MHz frequencies

Figure 4-4 shows the current distributions for frequencies 380 to 440 MHz. It can be seen that the magnitude of the current distribution increases with increasing frequency.

It can also be observed that the current on the largest diameter wire of the monopole antenna is higher than the current on the smallest diameter wire of the antenna when the excitation is on the middles wire. This is why the current values towards the end of the wire are higher after the sharp deceleration beyond the application of the excitation than the end of the wire before the sharp acceleration of the current values along the wire length.

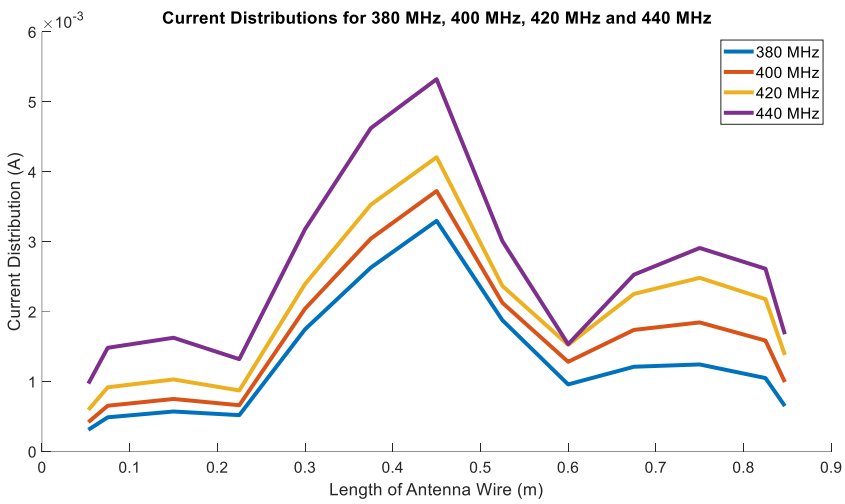


Figure 4-4: Current distributions for 380, 400, 420 and 440 MHz frequencies single excitation

Figure 4-5 shows that the antenna becomes more directional and beamforming with increasing frequencies as indicated by the gradual appearance of side lobes starting from the 400 MHz frequency.

The propagating distance continues to increase and the angular coverage continues to decrease with increasing frequencies. The higher directivity is obtained when $\theta = \pi/3$.

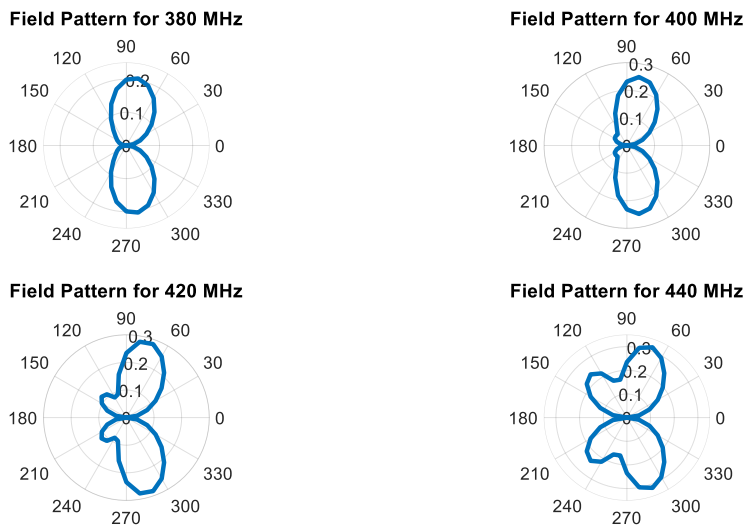


Figure 4-5: Field patterns for 380 to 440 MHz frequencies

Figure 4-6 shows that the maximum current distribution (7.83 mA) is obtained at 480 MHz at the point of excitation. At this point of excitation, frequencies below and above 480 MHz give lower value of current. Although, there are overlaps along the length of the antenna where the current values of 480 MHz frequency are less than currents for other frequencies.

At antenna length beyond 0.6 m, the current for the frequencies decreases with increasing frequency. Therefore, the current values for 460 MHz are higher than that of 480 MHz and so on. Also, the current values before the sharp acceleration of the current values are now higher than the current values after the sharp deceleration of current values beyond the excitation point.

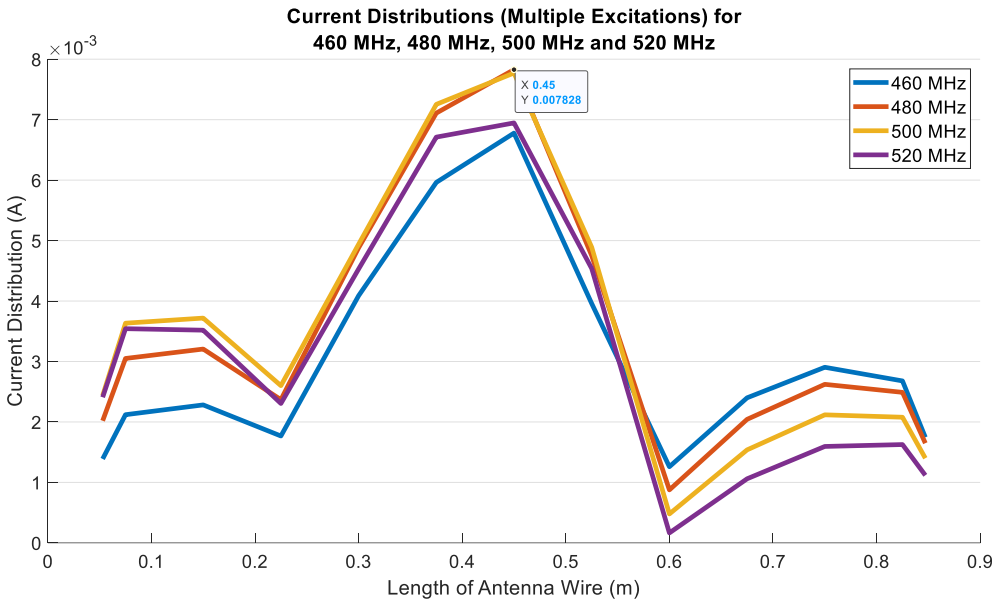


Figure 4-6: Current distributions for 460 to 520 MHz frequencies single excitation

Figure 4-7 shows that 460 MHz frequency is having two main lobes where the propagating distance of the appearing side lobes is approximately equal to the main lobe. At 480 - 520 MHz frequencies, the field patterns appeared to be more directional but in the opposite direction to 420 – 440 MHz frequencies and with higher propagating distance.

The previous main lobe completely disappeared at 480 MHz as the appearing side lobe becomes the main lobe and the angular coverage is tilted to the other side. The higher directivity is obtained at $\theta = 2\pi/3$.

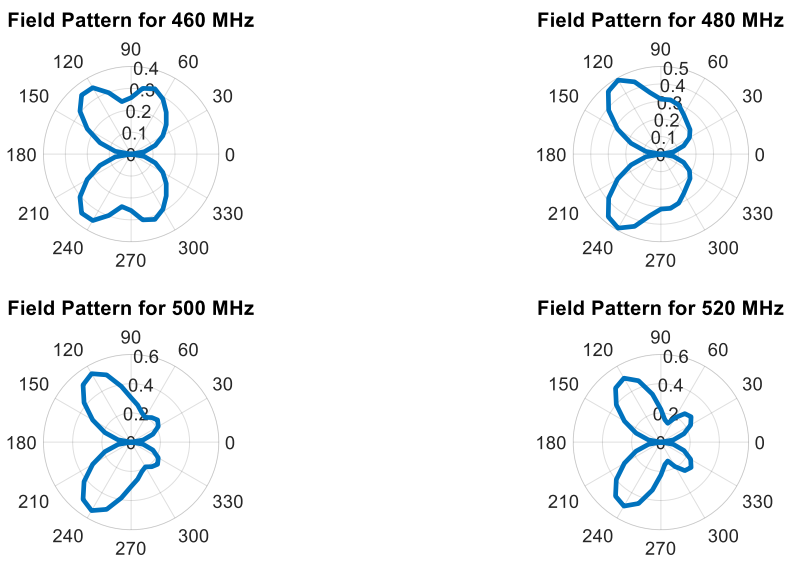


Figure 4-7: Field patterns for 460 to 520 MHz frequencies

Figure 4-8 shows that the magnitude of the current ditribution continue to decrease as the frequecies continue to increase beyond 480 MHz. The values of the current at the end of the smallest diameter wire of the antenna are higher than the wire with the largest diameter. This can be attributed to the exact point of excitation which is 0.4365 m of the antenna length

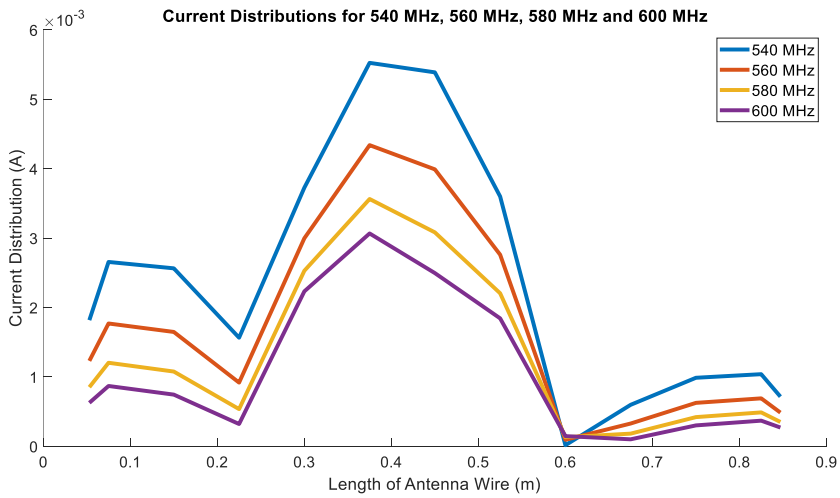


Figure 4-8: Current distributions for 540 - 600 MHz frequencies single excitation

A side lobe has begun apering from 520 MHz and it continues to grow with the propagating distance decreasing until we have almost two equal lobes as shown in figure 4-9 when the frequency is 600 MHz. It can be seen that the propagating distance through the centre is less than the two halves on either side for 600 MHz.

However, the angular coverage for 600 MHz is high. It can be seen that the higher directivity at 600 MHz occurred at both $\theta = \pi/2$ and $\theta = 2\pi/2$.

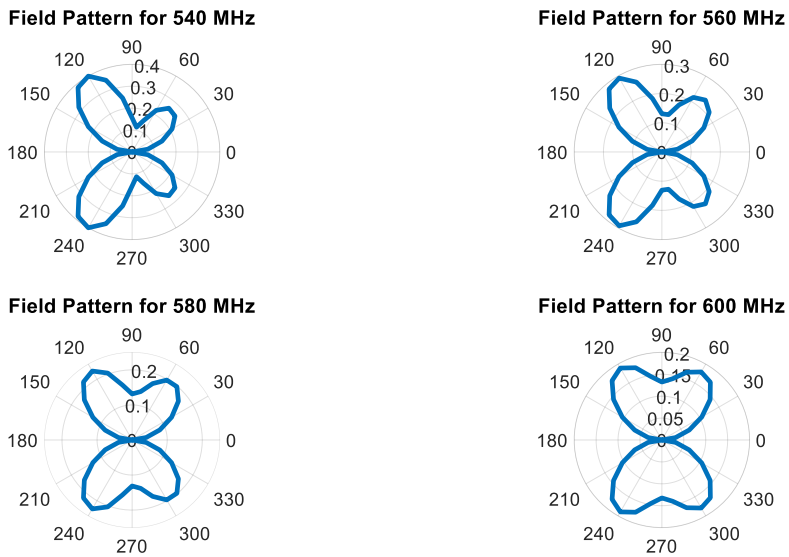


Figure 4-9: Field patterns for 540 – 600 MHz frequencies

It can then be concluded that if the antenna design goal is to achieve uniform radiating pattern, then the monopole antenna should be operated between 300 and 360 MHz. If angular coverage of the antenna is the goal with almost equal propagation distance on two halves of the antenna, then the antenna should be operated between 580 and 600 MHz.

However, if maximizing the propagating distance and current distribution magnitude are the goals, then the antenna should be operated between at 480 MHz.

Figure 4-10 shows the input impedance for the straight monopole antenna with single excitation.

Input impedance is defined as the impedance presented by an antenna at its terminals [49]. It can also be defined as the ratio of the voltage to current at a pair of terminals or the ratio of the appropriate components of the electric to magnetic fields at a point [10].

The input impedance of an antenna is generally a function of frequency. The input impedance continues to decrease as the frequency increases from 300 MHz until the minimum value (124.3 Ω) obtained at 480 MHz. Beyond this point, the input impedance steadily rises with increasing frequency. It can then be decuced that the current is inversely proportional to the input impedance, which statifies the Ohm's law.

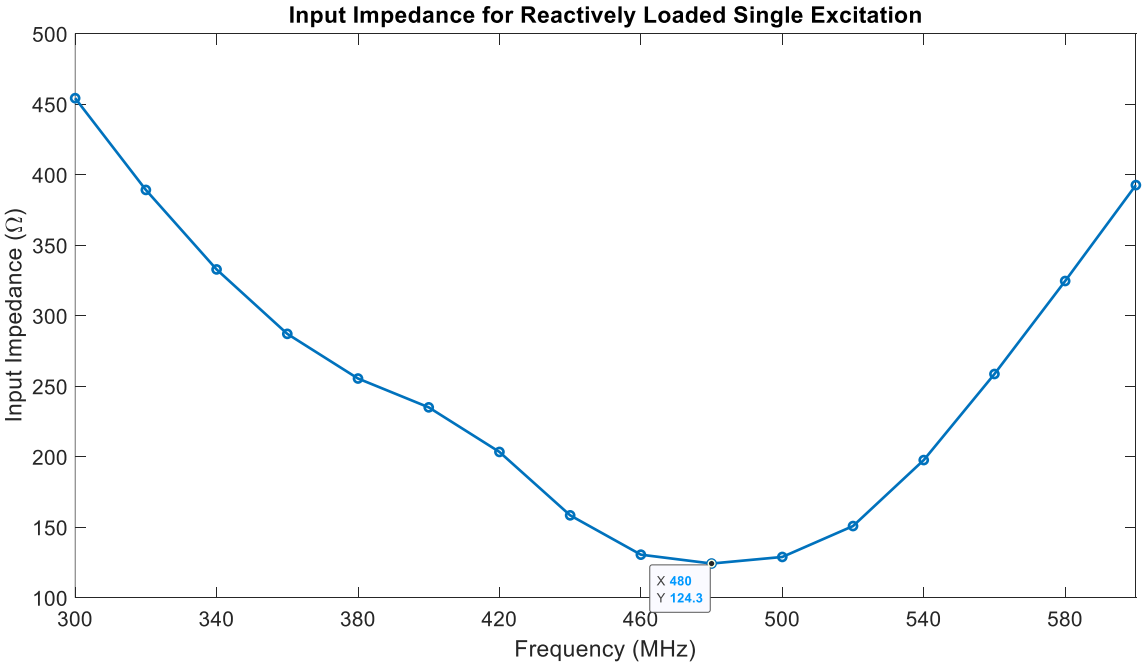


Figure 4-10: Input impedance for reactively loaded single excitation monopole straight antenna

The monopole wire antenna was then excited at multiple locations – 0.1455 m, 0.4365 m and 0.5955 m.

Figure 4-11 shows that the peak value of the current distribution magnitude for the same frequency range is higher for multiple excitation compared to single excitation.

The value of the current distribution increases with increasing frequency value. There are other peak points at the excitation locations. There is no third obvious peak for the frequency range 300 to 360 MHz.

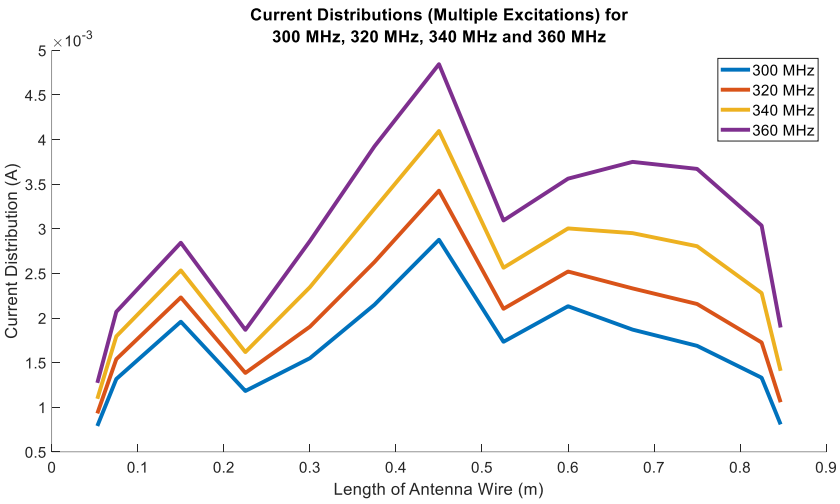


Figure 4-11: Current distributions for 300 - 360 MHz frequencies multiple excitation

Figure 4-12 shows that the propagation distance for multiple excitations is greater than that of the single excitation for the same frequency. For example, 360 MHz single excitation propataion distance is 0.2 while 360 MHz multiple excitation propagation distance is 0.45. The directivity is at $\theta = \pi/2$.

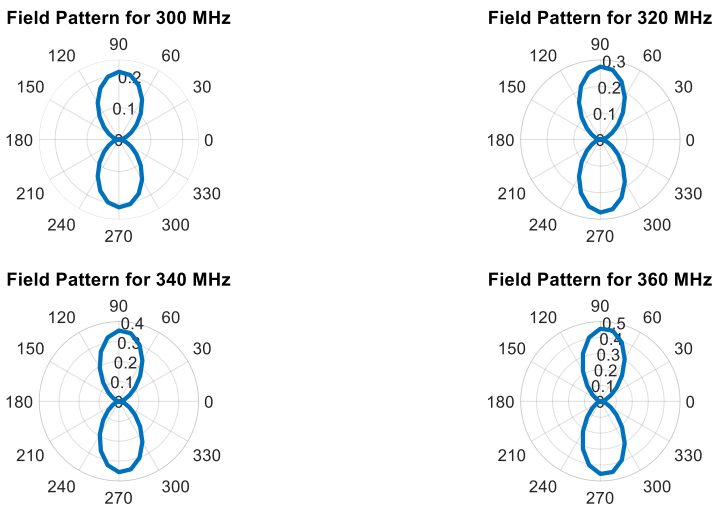


Figure 4-12: Field patterns for 300 – 360 MHz frequency range for multiple excitations

Figure 4-13 shows that the peak current magnitude continues to increase as the frequency increases. The third peak at the excitation point of 0.5955 m becomes obvious in the frequency range of 380 MHz and 440 MHz.

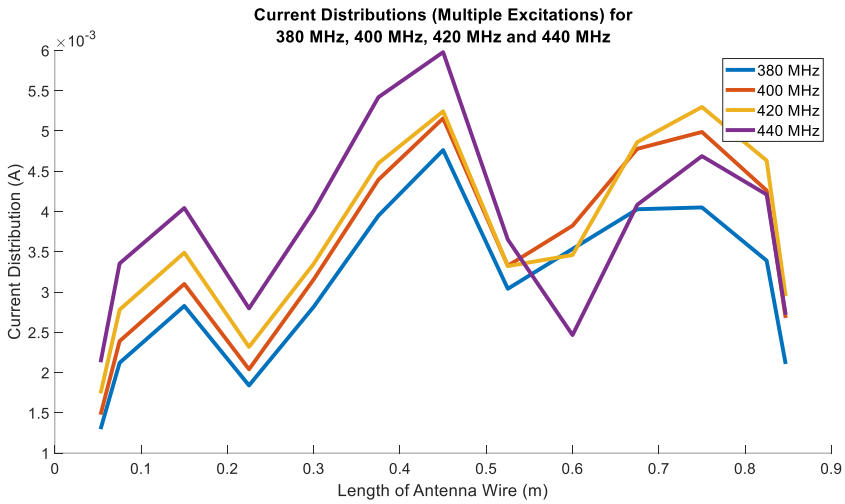


Figure 4-13: Current distributions for 380 - 440 MHz frequencies multiple excitation

Figure 4-14 shows a gradual appearance of side lobes from 420 MHz and the propagating distance is the same for frequencies 400, 420 and 440 MHz. θ corresponding to the directivity continues to shift away from $\pi/2$ to $\pi/3$. At 440 MHz, there is a visible sidelobe.

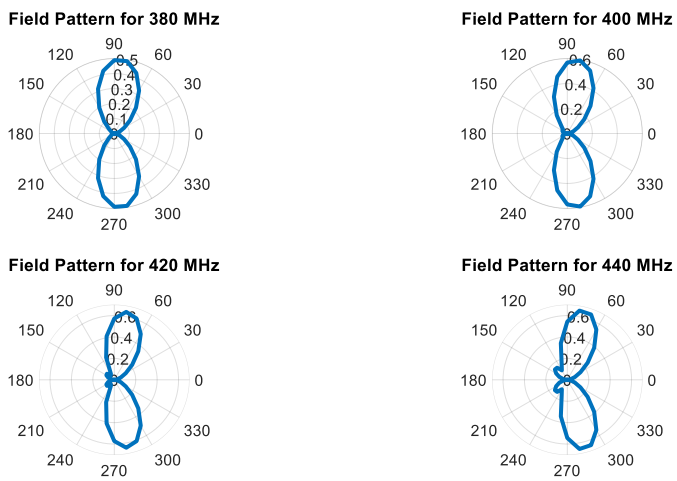


Figure 4-14: Field patterns for 380 – 440 MHz for multiple excitations

Figure 4-15 shows the maximum current distribution is obtained at 480 MHz. Frequencies below and above 480 MHz gives lower current distribution value. There is a considerable difference when the maximum current is achieved at 480 MHz and the current of 500 MHz. The maximum current value is 7.88 mA which is greater than 7.83 mA for single excitation.

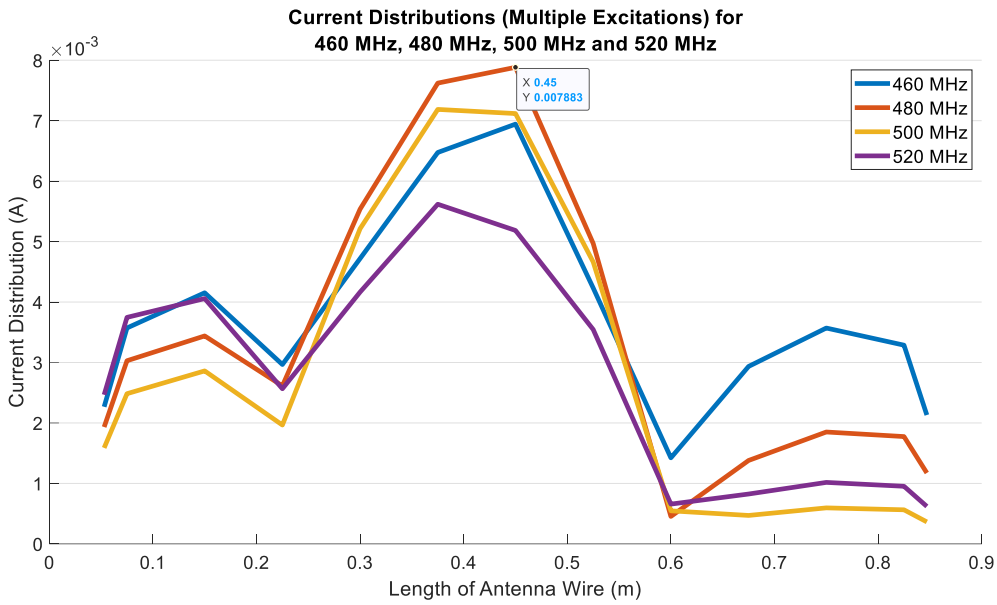


Figure 4-15: Current distributions for 460 - 520 MHz frequencies multiple excitation

Figure 4-16 shows that the side lobes that started appearing at 420 MHz is disappearing from 460 MHz. Another observation is that the angular coverage decreases as the frequency increases

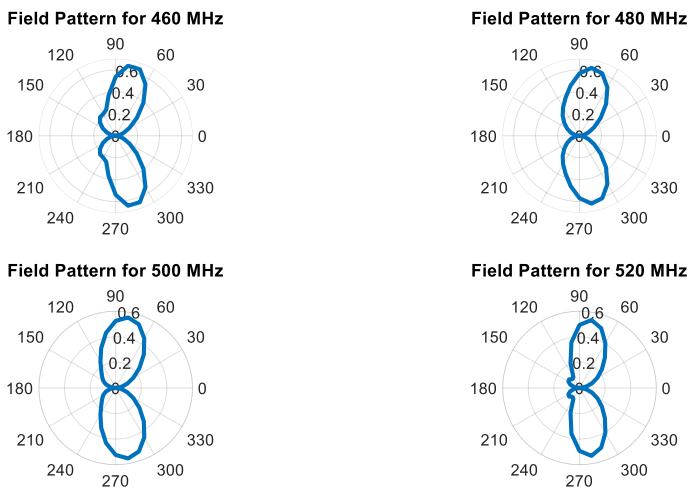


Figure 4-16: Field patterns for 460 to 520 MHz frequency range

Figure 4-17 shows that the peak current magnitude corresponds to excitation at one end of the antenna while the value of the current magnitude of the centre excitation continue to deccrease with increasing frequency. The frequency range shows that the current at the first excitation point is now higher than the excitation in the middle of the antenna.

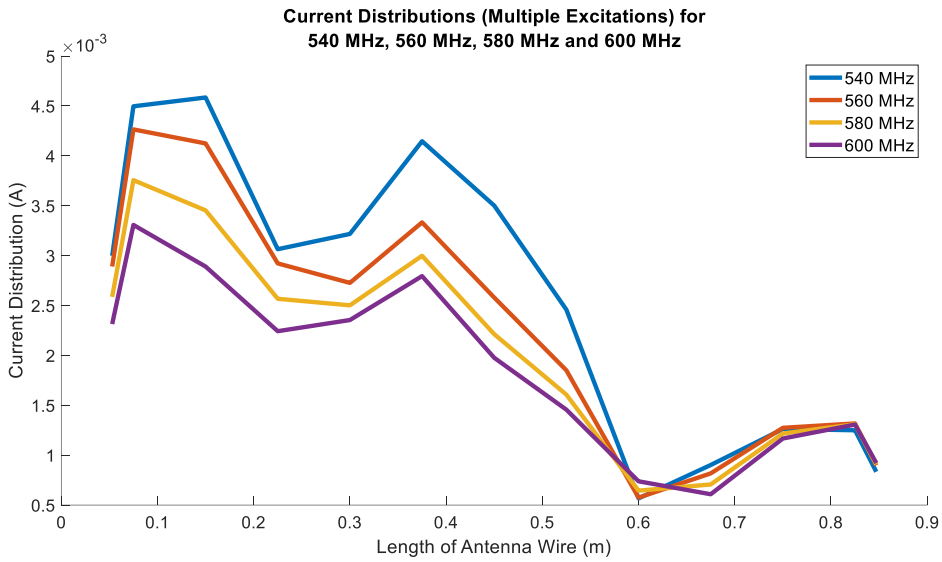


Figure 4-17: Current distributions for 540 - 600 MHz frequencies multiple excitation

Figure 4-18 shows that the angular coverage and the propagation distance continues to decrease as the frequency increases. Also, the side lobe completely disappeared. It can be seen that the directivity is at $\theta = \pi/2$.

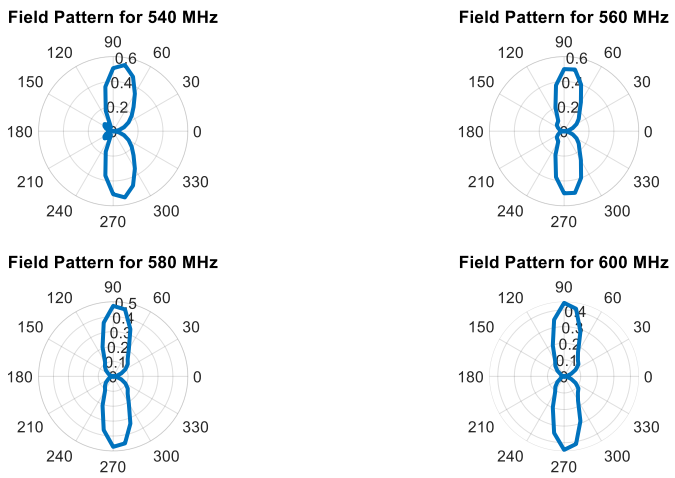


Figure 4-18: Field patterns for 540 – 600 MHz multiple excitations

It can then be concluded that if higher propagation distance is the goal, then multiple excitation is better than a single excitation for the same frequency. However, if the angular coverage maximisation is the goal, then single excitation is a better choice for this monopole antenna.

Figure 4-19 shows the input impedance for multiple excitations for the monopole antenna that is reactively loaded. Unlike the singly excited case where the lowest input impedance value corresponds to the highest current, the highest input impedance corresponds to the highest

current in multiple excitation with 2,040 Ω. The lowest input impedance is obtained at 400 MHz with a value of 246.1 Ω.

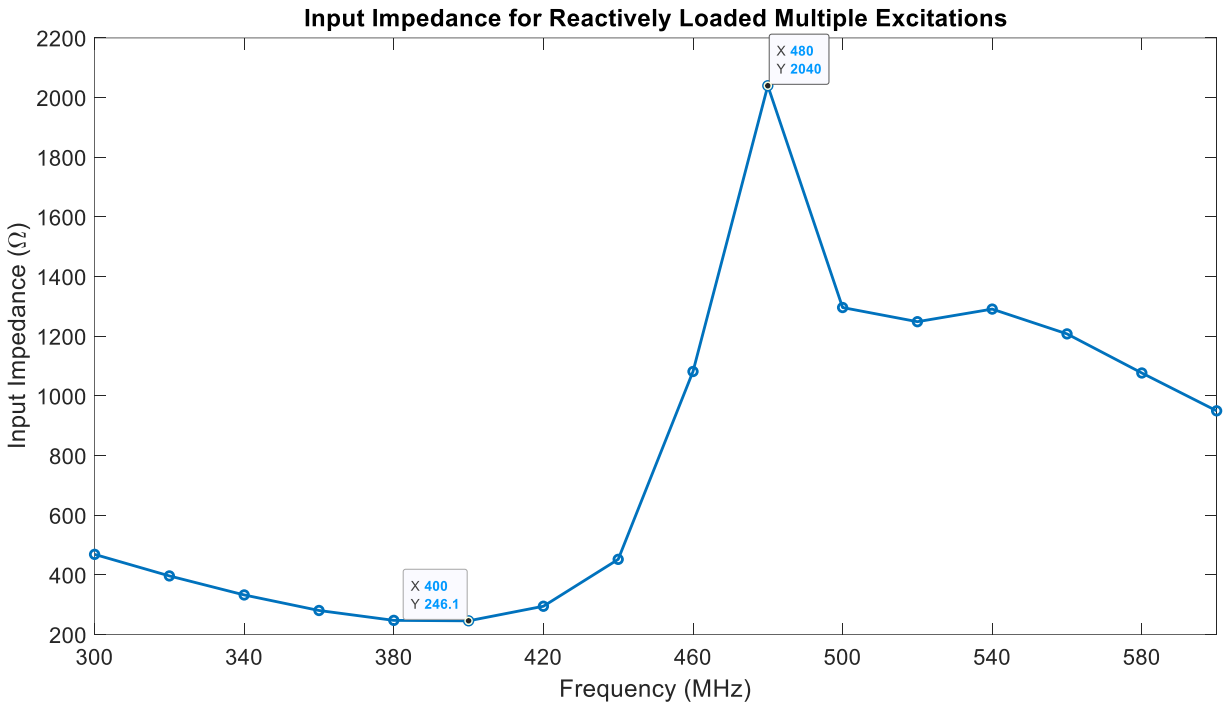


Figure 4-19: Input impedance of multiple excitations for reactively loaded straight monopole antenna

4.2. Scenario 2 – Resistively Loaded

This scenario models resistively loading the wire antenna with 100 Ω at the centre of the wire. Figure 4-18 shows the current distribution of the wire antenna when it is resistively loaded at the centre. Similar to the case of reactively loaded monopole antenna above, the magnitude of the current distribution continues to increase as the frequency increases.

However, the current distribution values of the reactively loaded monopole antenna are greater than that of the resistively loaded antenna for the same frequency.

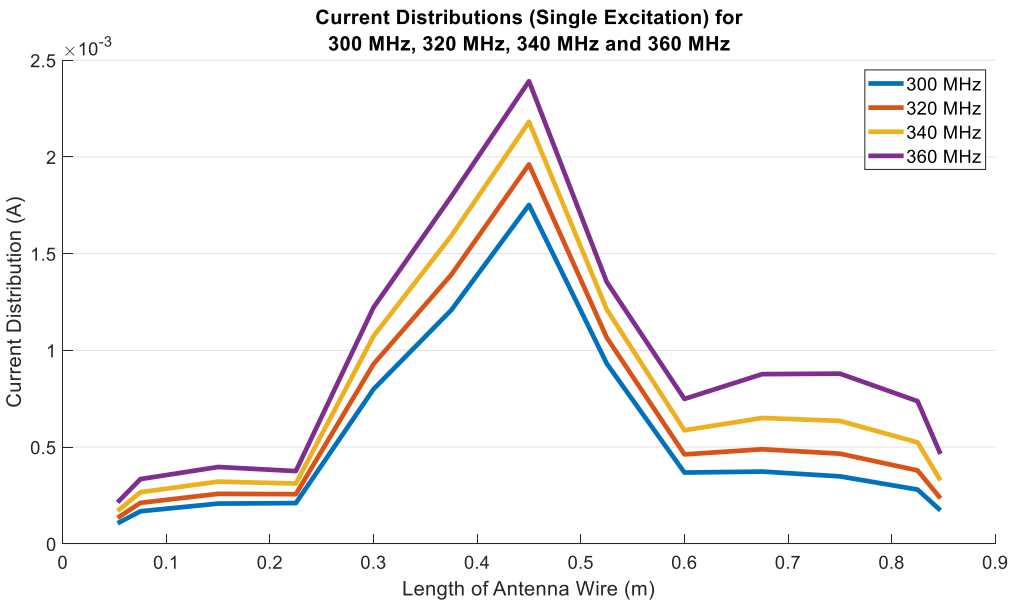


Figure 4-20: Current distributions for 300, 320, 340 and 360 MHz frequencies single excitation

The propagation distance continues to increase as the frequency increases for the frequency range of 300 to 360 MHz. The directivity value of the antenna when resistively loaded is smaller than when reactively loaded for the same frequency. Similar to the reactively loaded scenario, the directivity is at $\theta = \pi/2$

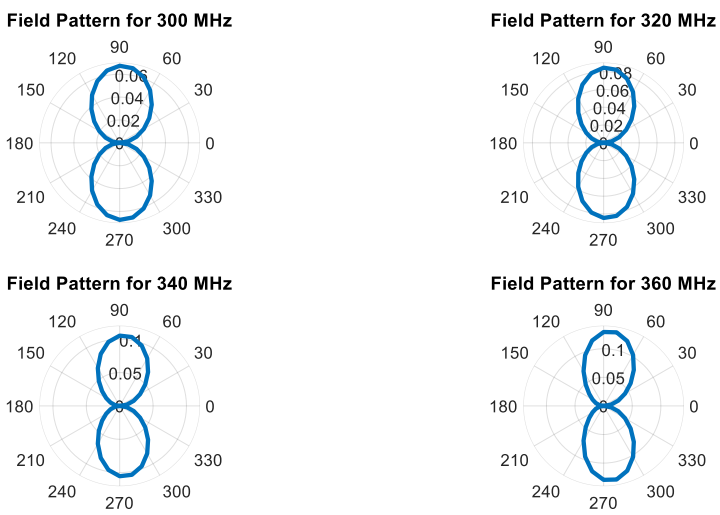


Figure 4-21: Field patterns for 300 – 360 MHz

Figure 4-22 shows that the trend of increasing current distribution with increasing frequencies continues. The peak current at 400 MHz is about 3.2 mA for resistively loaded and 5.2 mA reactively loaded

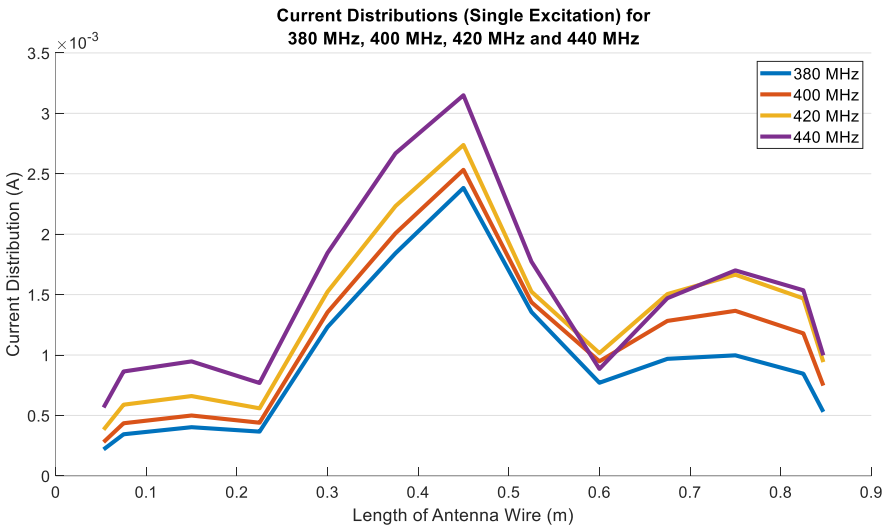


Figure 4-22: Current distributions for 380 - 440 MHz frequencies single excitation

Side lobes gradual appearane is observed from 420 MHz in figure 4-21. At 440 MHz, the higher directivity is 0.2 when it is resistively loaded and 0.3 when it is reactively loaded at $\theta = \pi/3$. The other prominent direction of the radiation is at $\theta = 2\pi/3$ for the 440 MHz for the resistively loaded and the reactively loaded scenarios.

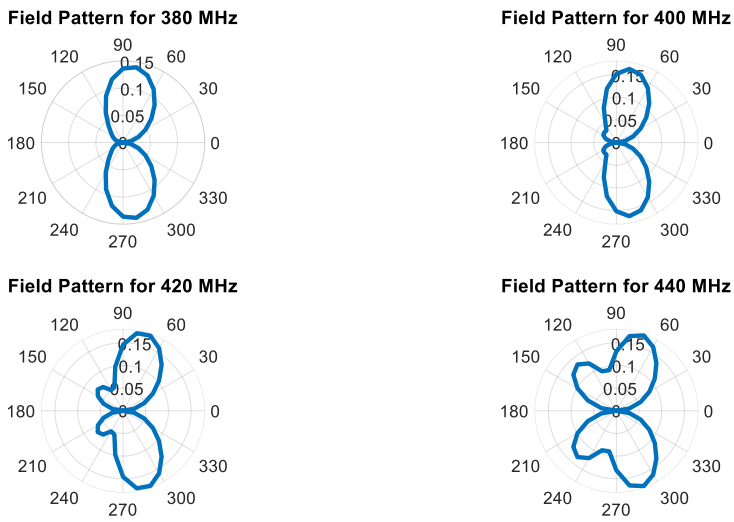


Figure 4-23: Field pattern for 380 – 440 MHz frequency range

Figure 4-24 shows that unlike reactively loaded that reaches its maximum current of 7.83 mA at 480 MHz, maximum current of 4.16 mA is obtained at 520 MHz for the resistively loaded scenario.

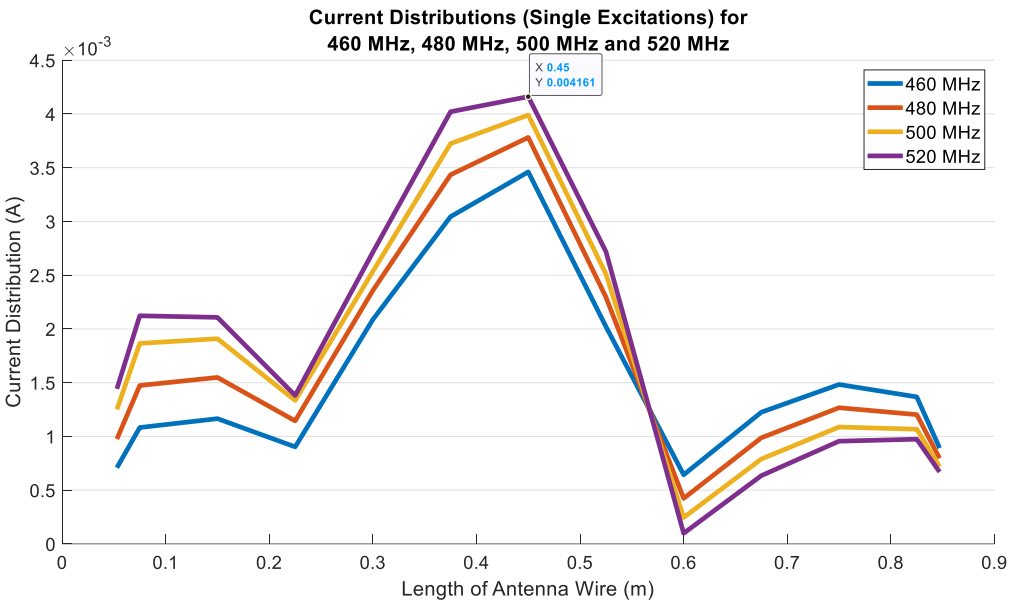


Figure 4-24: Current distributions for 460 - 520 MHz frequencies single excitation

The side lobe is of equal propagation distance with the main lobe at 460 MHz and transitions to be the main lobe from 480 MHz while the propagation distance continues to increase as shown in figure 4-23. There are two directivities at 460 MHz with 0.2 at $\theta = 2\pi/3$ and 0.15 at $\theta = \pi/3$. The higher directivity is at $\theta = 2\pi/3$.

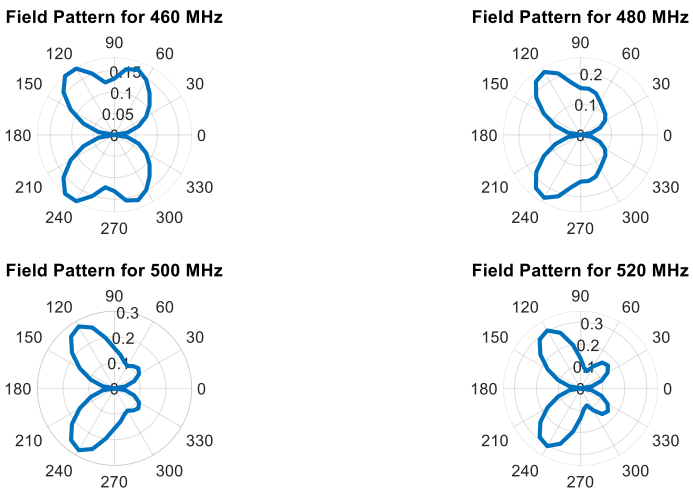


Figure 4-25: Field patterns for 460 – 520 MHz

Figure 4-26 shows that the maximum current distribution for single excitation when the antenna is resistively loaded is between 520 and 540 MHz. Frequencies below or above 520 MHz have their peak current values lower than that of the 520 MHz frequency as shown for the frequency range 540 and 600 MHz.

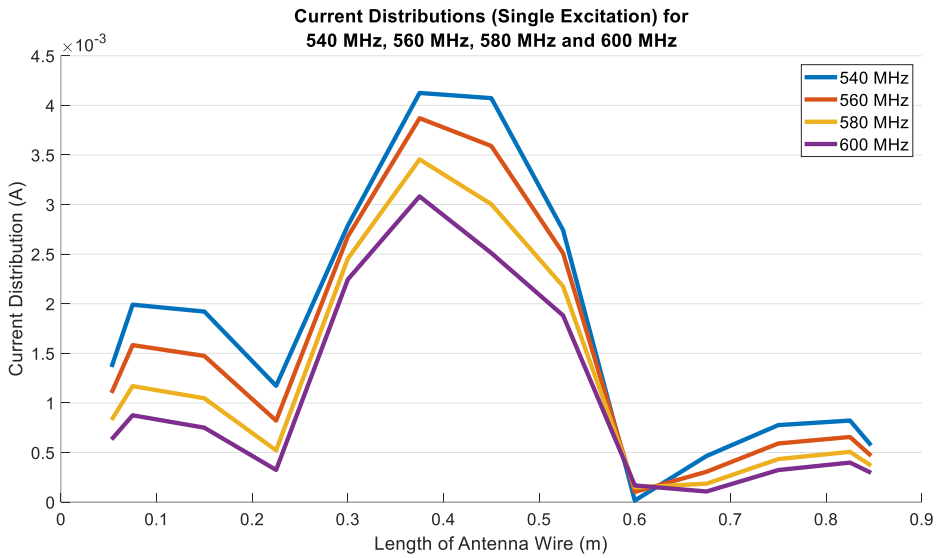


Figure 4-26: Current distributions for 540 - 600 MHz frequencies single excitation

Figure 4-27 show that the propagation distance of 540 MHz is the highest and the same as that of 520 MHz. However, 540 MHz has more prominent sidelobe tha 520 MHz frequency. Similar to reactively loaded, the field patterns have equal halves either side of the centre of the antenna

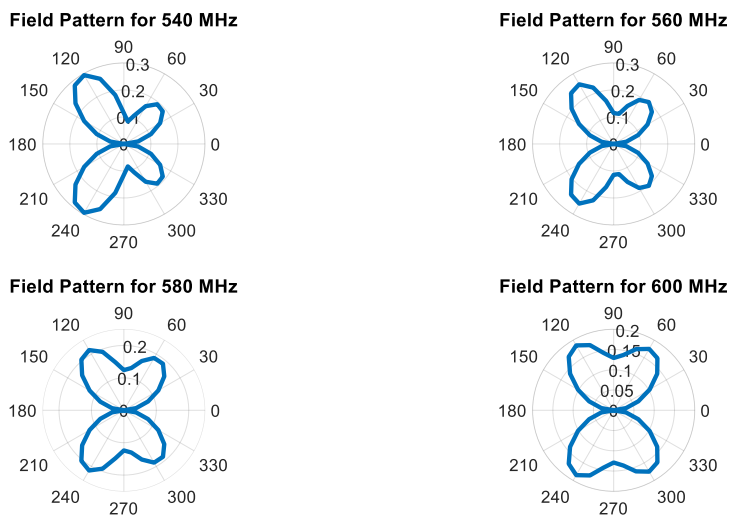


Figure 4-27: Field patterns for 540 – 600 MHz

Figure 4-28 shows that the current distribution value for multiple excitation is higher than the single excitation for the same frequencies.

Figure 4-29 shows the input impedance of straight monopole antenna that is resistively loaded and excited at a single position by a unit voltage. The minimum input impedance of 237.1 Ω is at 520 MHz, which is the same frequency where the maximum current of 4.16 mA was

obtained. The minimum input impedance for resistively loaded is higher than that of the reactively loaded of 124.3 Ω.

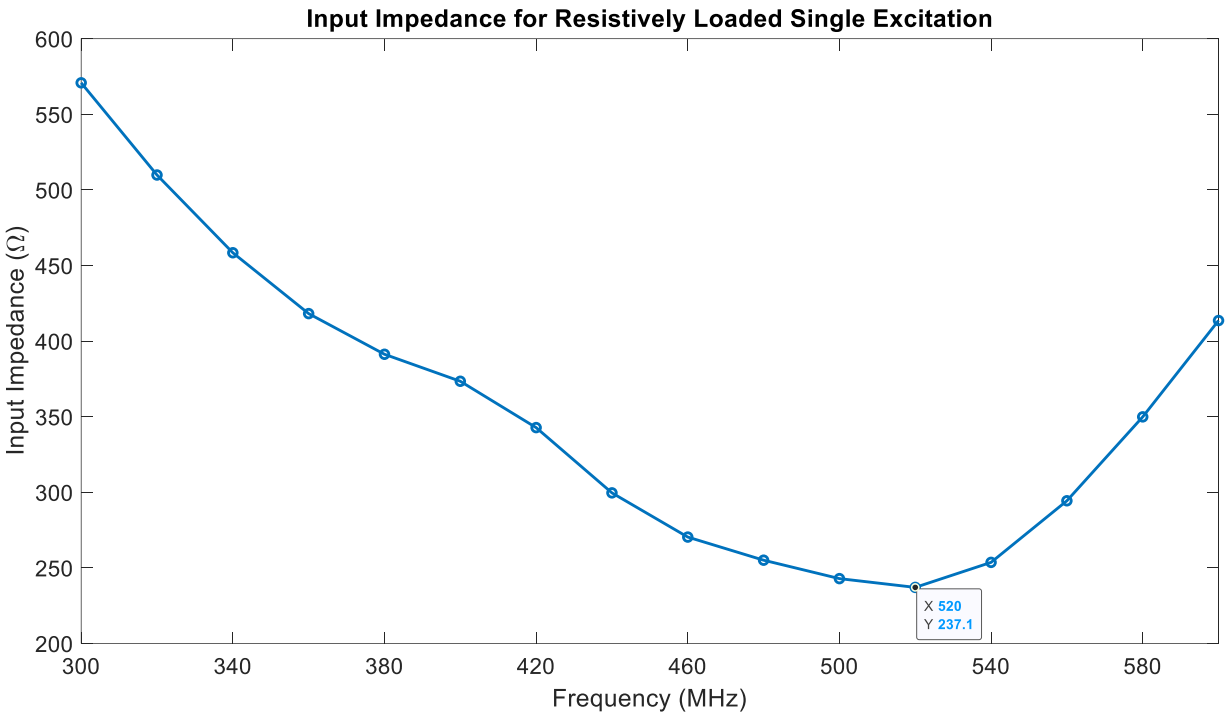


Figure 4-28: Input impedance of resistively loaded monopole antenna singly excited

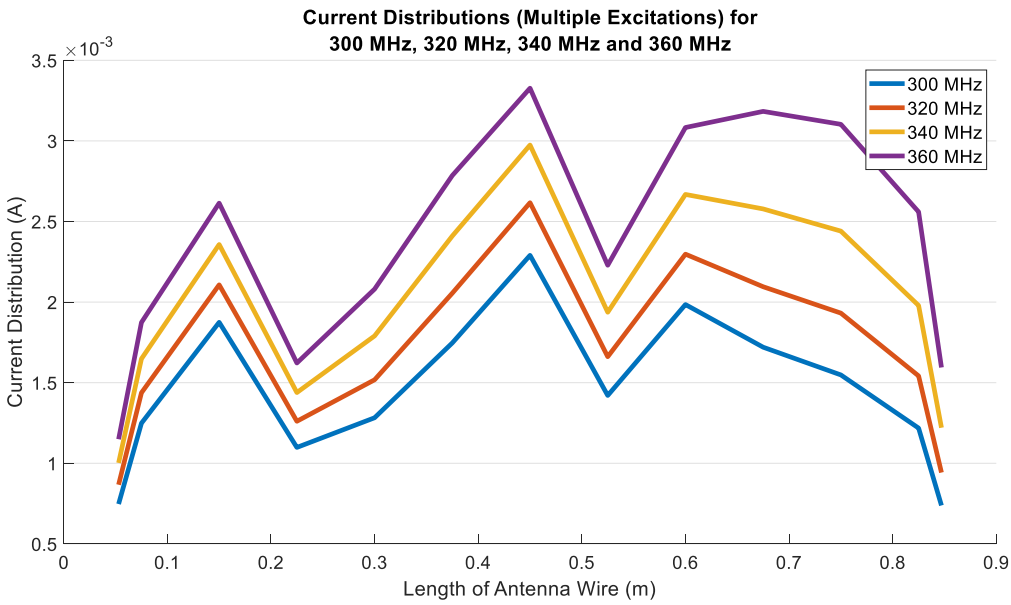


Figure 4-29: Current distributions for 300 - 360 MHz frequencies multiple excitation

The corresponding propagation distances for multile excitations are higher than the single excitation as shown in figure 4-27.

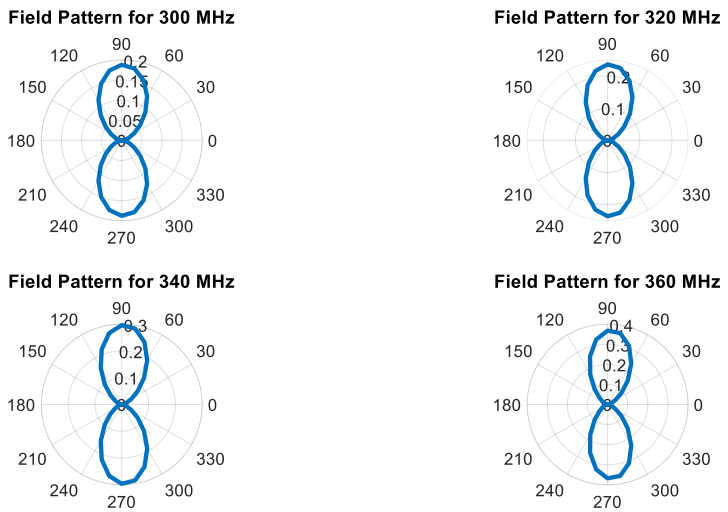


Figure 4-30: Field patterns for 300 – 360 Mhz

Figure 4-31 shows that the current distribution values of the two excitations at the two ends of the straight monopole antenna are higher than the excitation at the centre. This is unlike the reactively loaded scenario where peak current at the middle excitation point of the antenna is greater than the currents at the other two ends of the antenna.

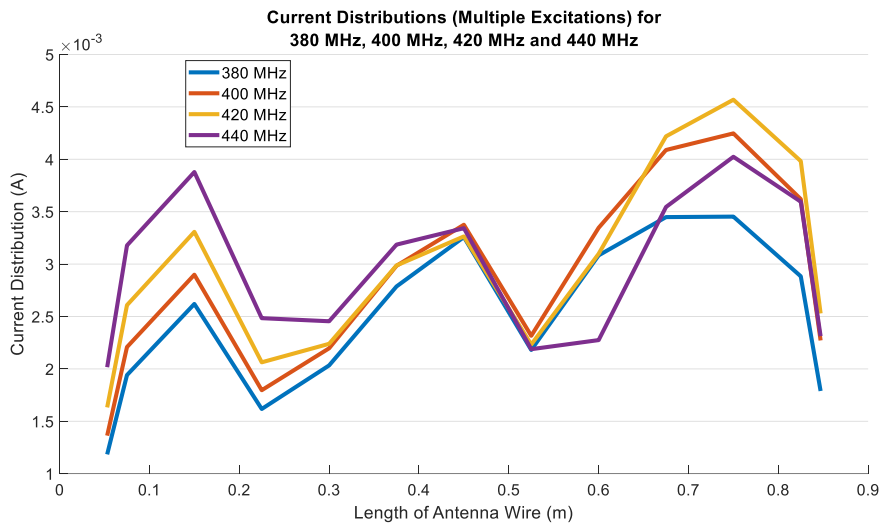


Figure 4-31: Current distributions for 380 - 440 MHz frequencies multiple excitation

The propagation distance continues to increase as the freuqncy increases with the gradual appearance of side lobes at 420 MHz

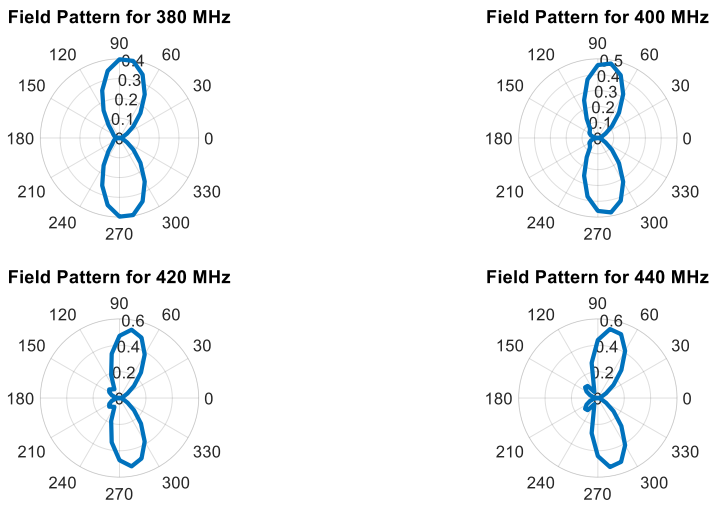


Figure 4-32: Field patterns for 380 – 440 MHz

Figure 4-33 shows that the current distribution continues to increase with increasing frequencies on one end of the excitation while it decreases with increasing frequencies on the other end of the excitations for frequency range 460 – 520 MHz. The maximum current is 4.9 mA at the lower end of the antenna while that of the single excitation is 4.16 mA at the same frequency.

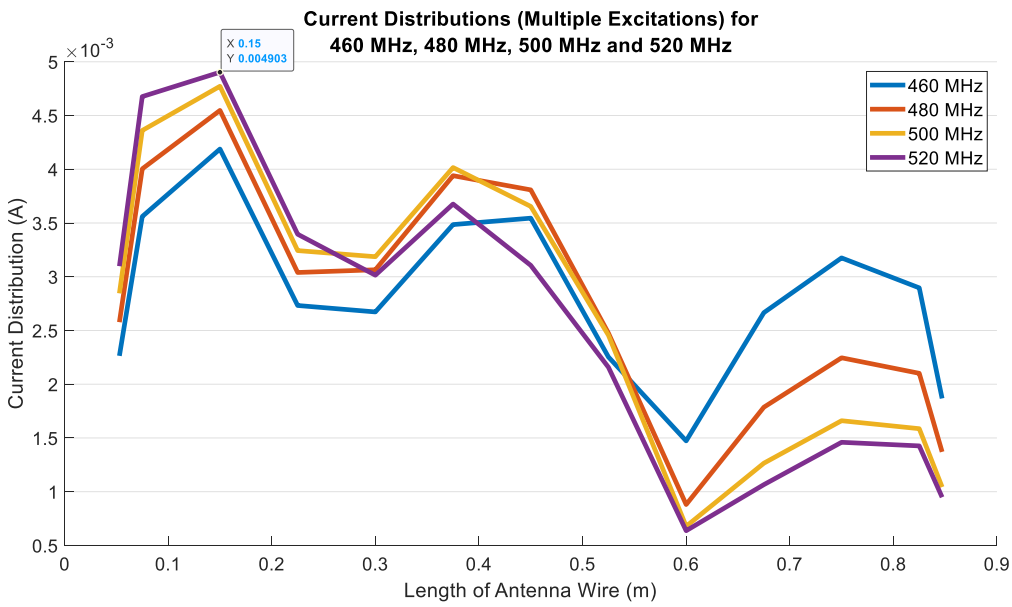


Figure 4-33: Current distributions for 460 - 520 MHz frequencies multiple excitation

The sidelobe which started appearing at 420 MHz disappeared at 480 MHz

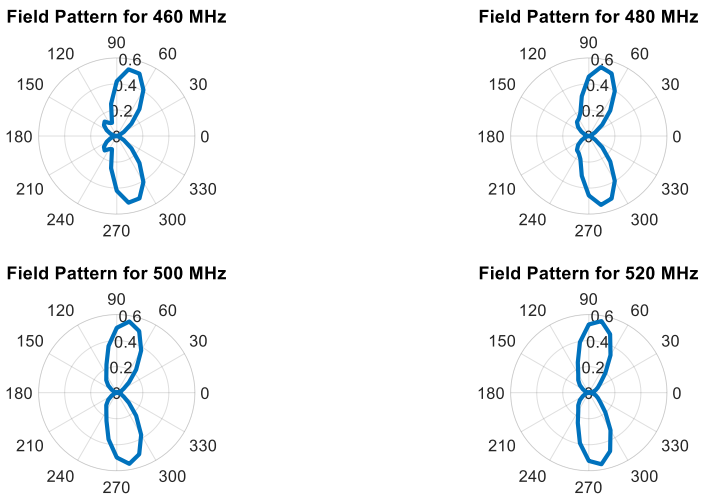


Figure 4-34: Field patterns for 460 – 520 MHz

Figure 4-35 shows that the current distribtion continues to derease beyond 540 MHz at the point of the three excitations

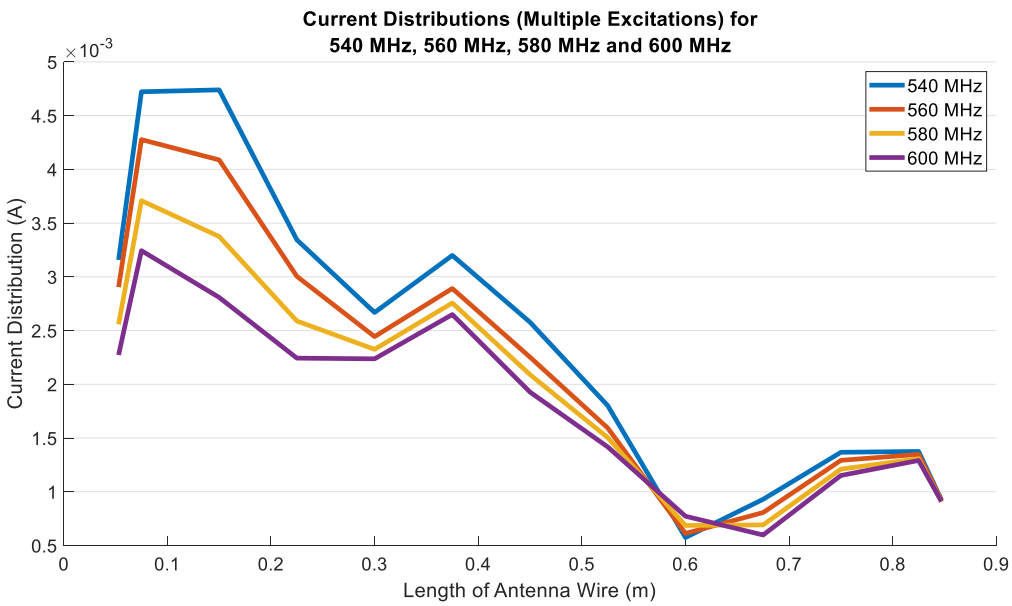


Figure 4-35: Current distributions for 540 - 600 MHz frequencies multiple excitation

The propagation distance continues to decrease with increasing frequencies with no side lobes as shown in figure 4-36.

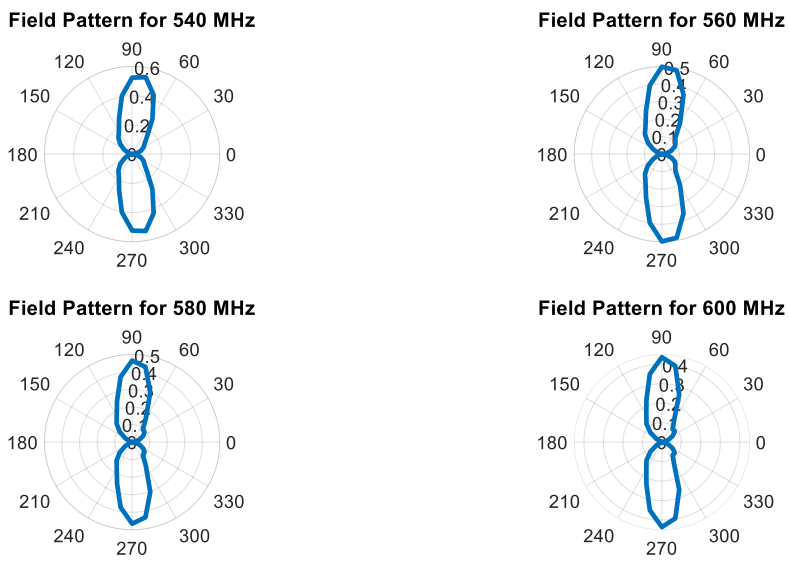


Figure 4-36: Field patterns for 540 – 600 MHz

Figure 4-37 shows that the minimum input impedance of 282.2 Ω is at 400 MHz similar to that of the reactively loaded scenario but with a value of 246.1 Ω. While the maximum input impedance of reactively loaded scenario is 2,040 Ω at 480 MHz, the maximum input impedance for the resistively loaded scenario is 1,246 Ω at 520 MHz.

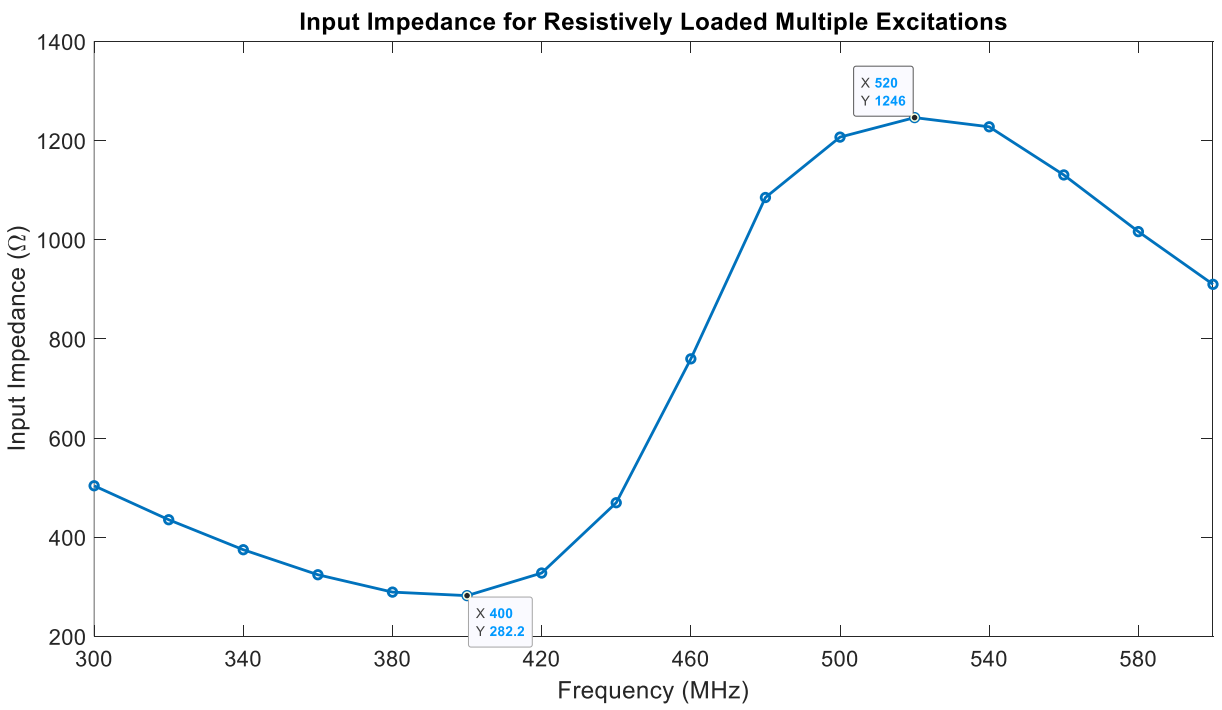


Figure 4-37: Input impedance of resistively loaded monopole antenna with multiple excitation

4.4. Scenario 3 – Resistively and Reactively Loaded

In this scenario, the wire antenna is resistively and reactively loaded with 100+j100 Ω. Figure 4-34 shows that the magnitudes of the current distribution for the frequency range between 300 and 360 MHz are greater than resistively loaded antenna but less than reatively loaded antenna. It follows the trend of increasing current distribution with inreasing frequency.

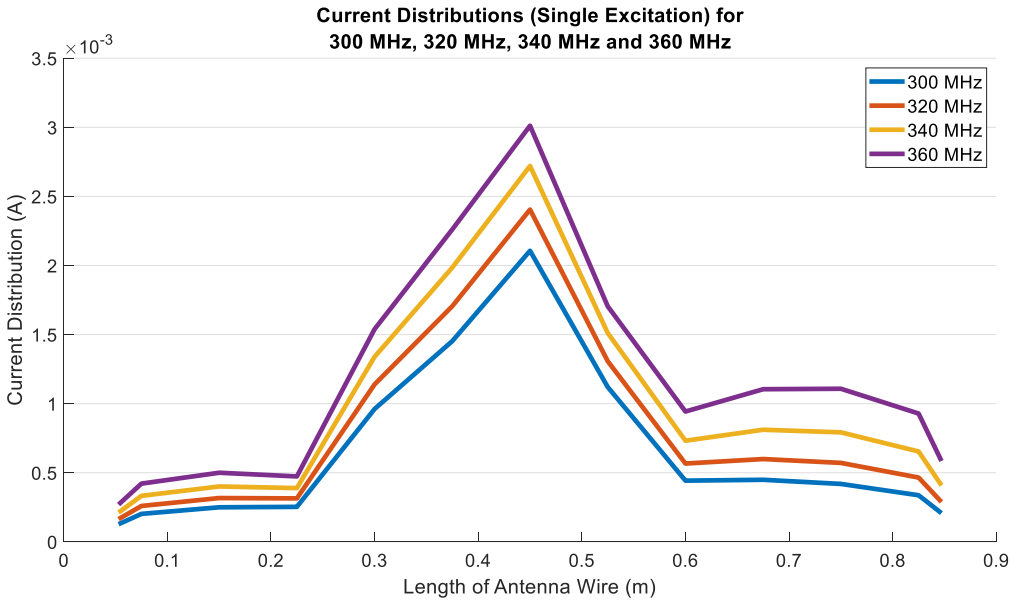


Figure 4-38: Current distributions for 300 - 360 MHz frequencies single excitation

The propagation distance increases with increasing frequency for the range between 300 and 360 MHz s shhwon in figure 4-39.

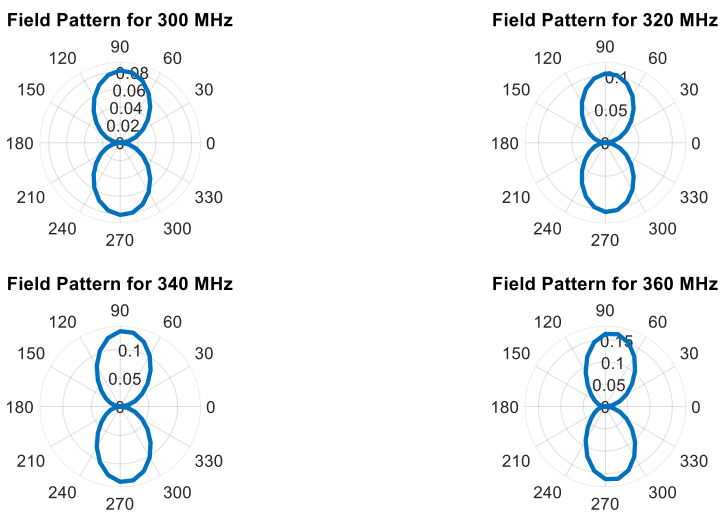


Figure 4-39: Field patterns for 300 – 360 MHz

Figure 4-40 shows that the current distribution continues to increase with increasing frequency.

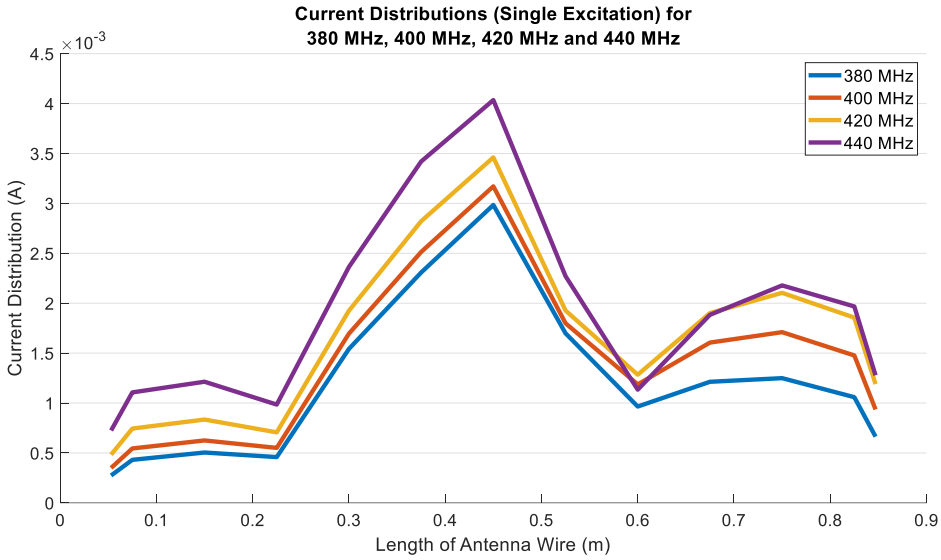


Figure 4-40: Current distributions for 380 - 440 MHz frequencies single excitation

There is a gradual appearance of side lobe from 420 MHz and the propagation distance continues to increase with increasing frequency.

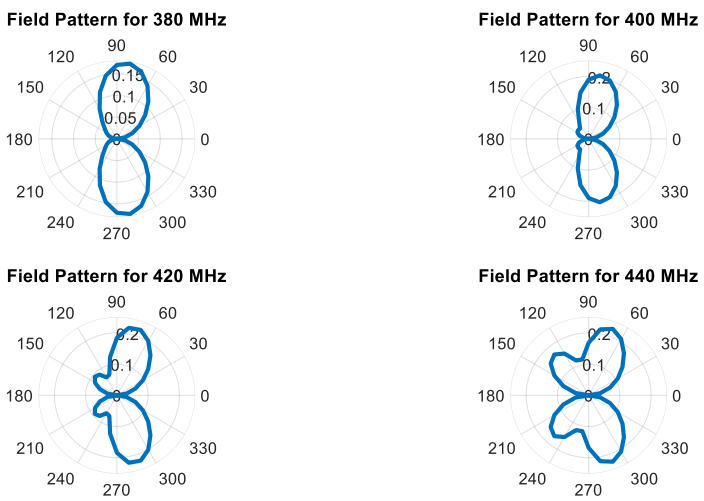


Figure 4-41: Field patterns for 380 – 440 MHz

The maximum crrent distribution is at 480 MHz. Frequencies below an above 480 MHz have lower current distribution value.

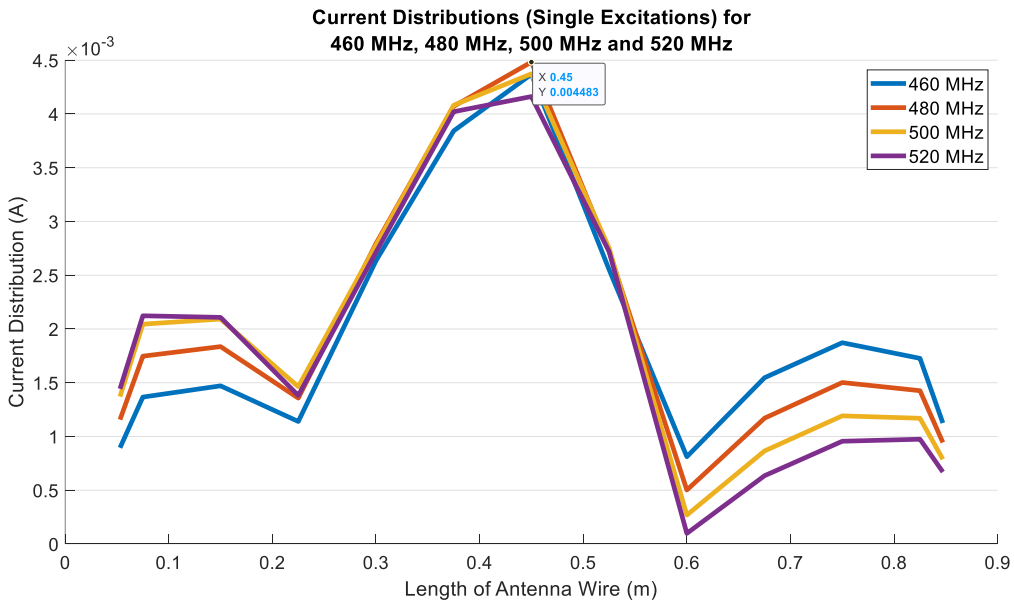


Figure 4-42: Current distributions for 460- 520 MHz frequencies single excitation

The propagation distances of 500 and 520 MHz are greater than 480 MHz. However, both 500 and 520 have side lobes while there is no side lobe for 480 MHz frequency.

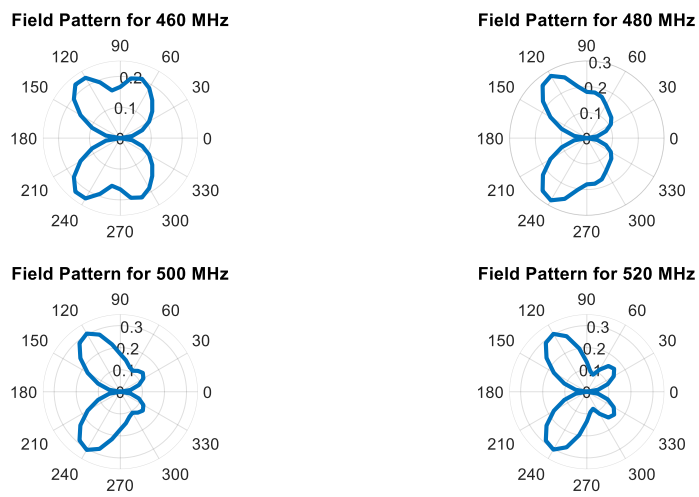


Figure 4-43: Field patterns for 460 – 520 MHz

The current distribution for the frequency range 540 to 600 MHz shows that the magnitude of the current frequency continues to decrease as the frequency increases.

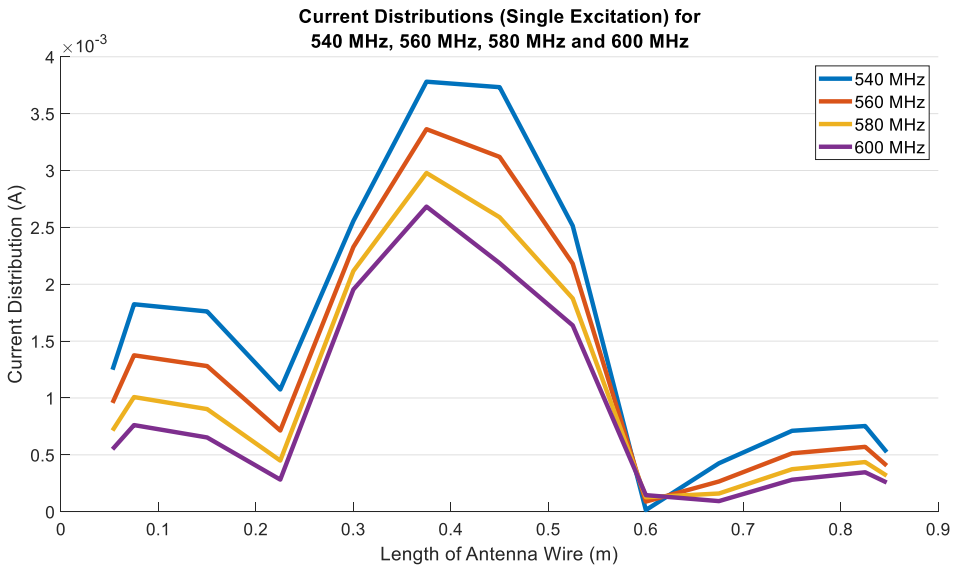


Figure 4-44: Current distributions for 540 - 600 MHz frequencies single excitation

Beyond 520 MHz, the propagation distance continues to decrease until the side lobe is fully developed as the main lobe when the frequency is 600 MHz.

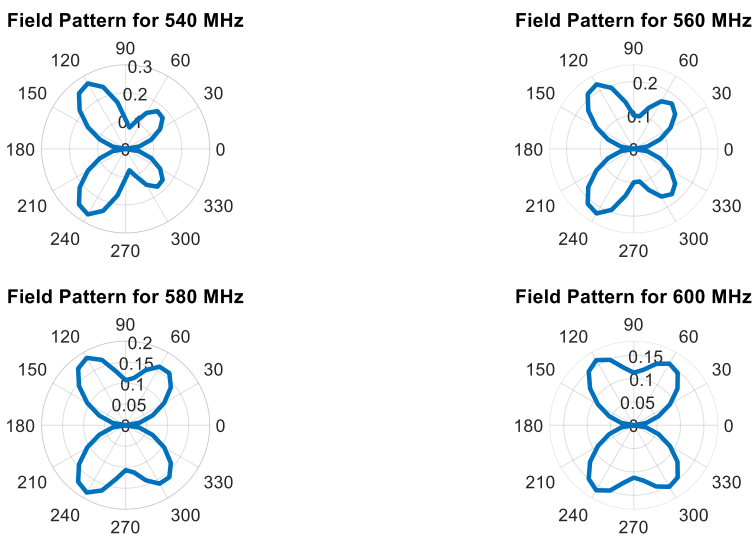
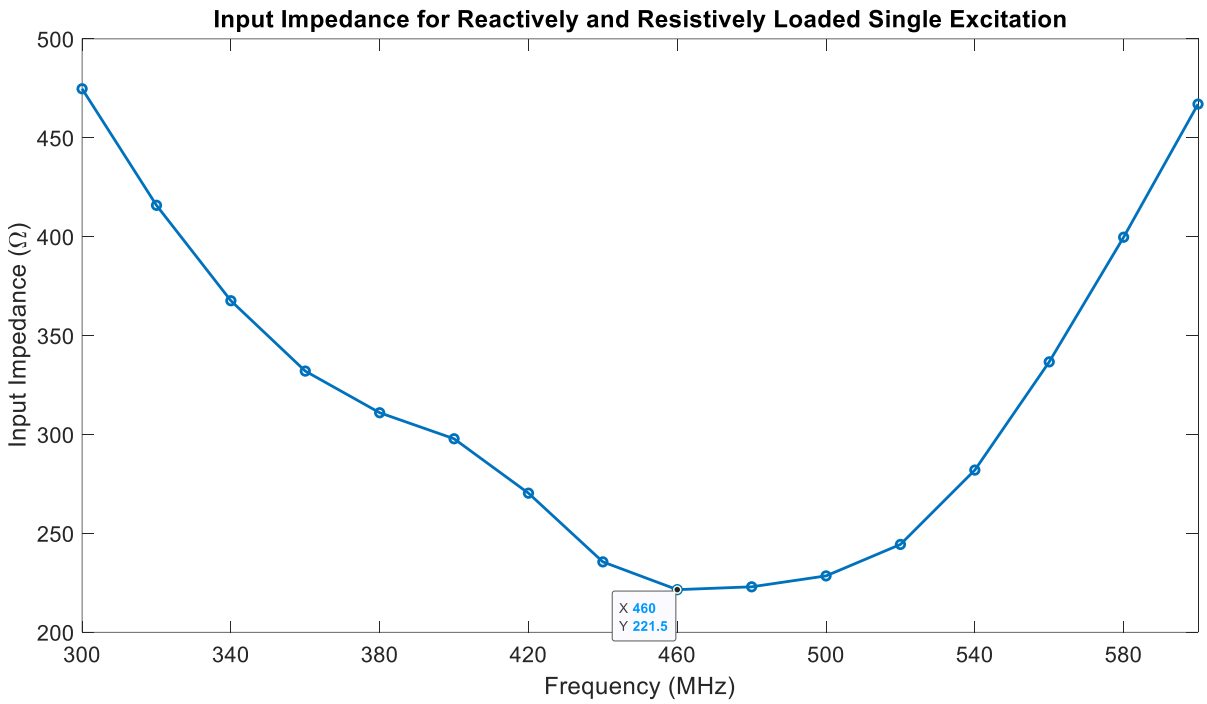


Figure 4-45: Field patterns for 540 – 600 MHz

Figure 4-46 shows that the magnitude of the crrent distribution for multiple excitation is greater than the single excitation for a monopole antenna that is resistively and reactively loaded. The current distribution increases with increasing frequency.



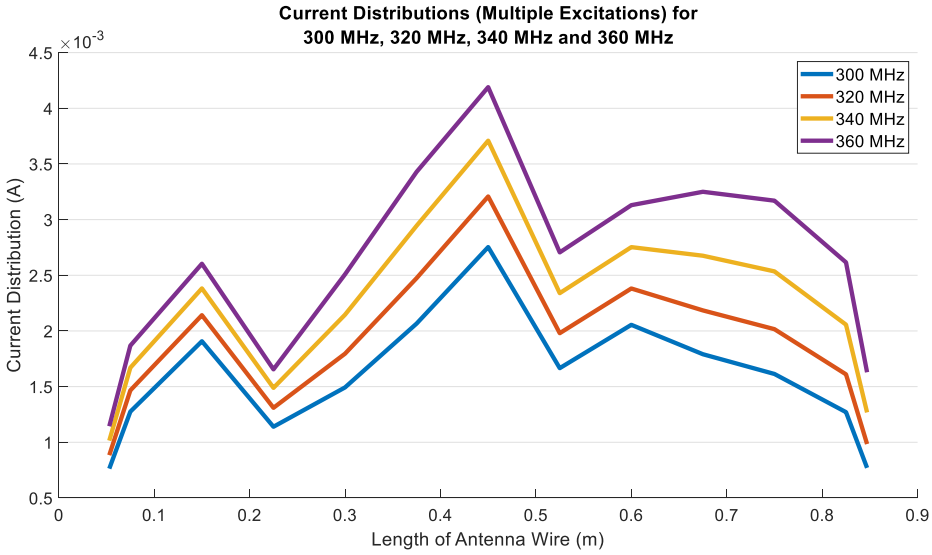


Figure 4-46: Current distributions for 300 - 360 MHz frequencies multiple excitation

The propagation distances increases and the angular coverage decreases with increasing frequency for the frequency range between 300 and 360 MHz as shown in figure 4-47.

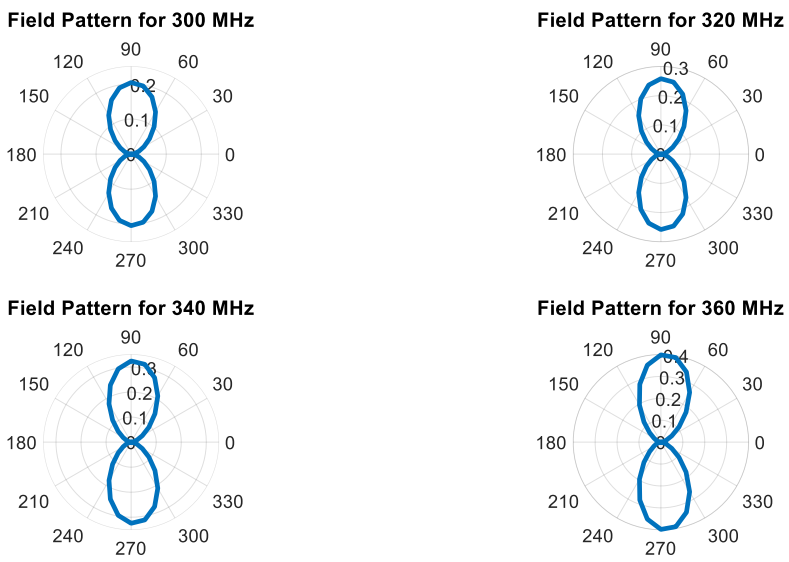


Figure 4-47: Field patterns for 300 – 360 MHz

The current distribution corresponding to one side of the antenna increases rapidly to almost similar values of the excitation in the middle of the antenna for the frequency range between 380 and 440 MHz as shown in figure 4-48.

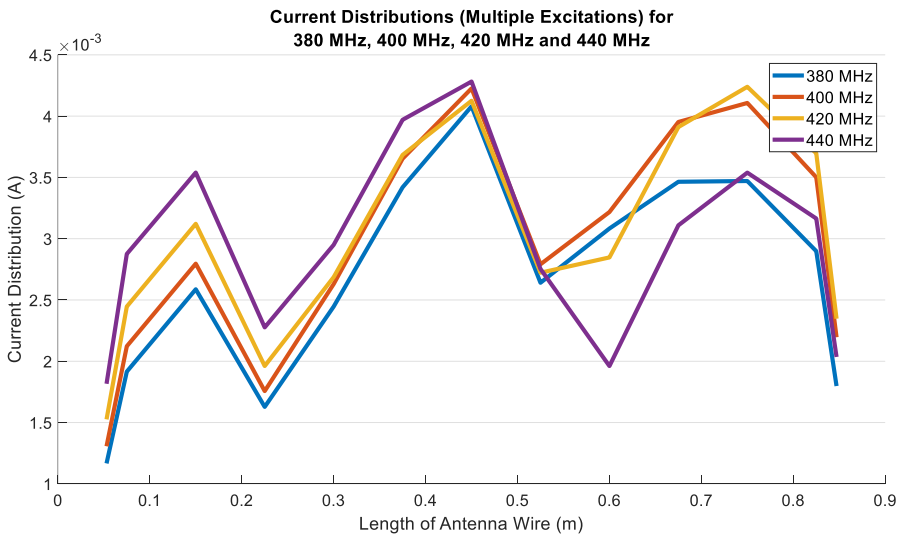


Figure 4-48: Current distributions for 380 – 440 MHz frequencies multiple excitation

Side lobes gradual appearance was observed at 440 MHz as shown in igure 4-49.

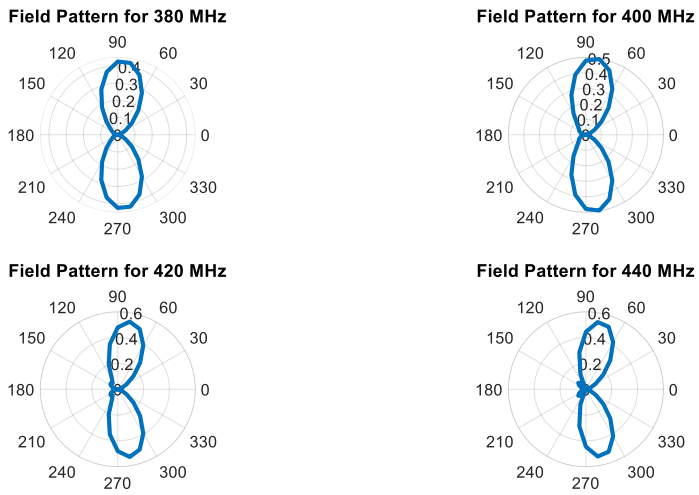


Figure 4-49: Field patterns for 380 – 440 MHz

The current distribution continues to increase with increasing frequency on one end of the antenna where it was excited as shown in figure 4-50.

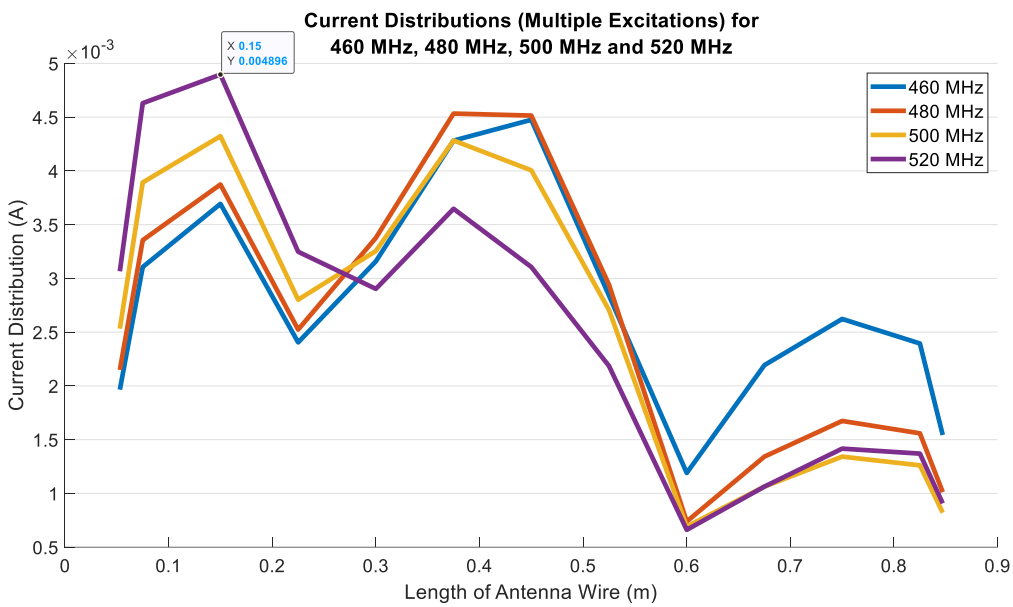


Figure 4-49: Current distributions for 460 – 520 MHz frequencies multiple excitation

The side lobes which started appearing at 480 MHz frequency have completely disappeared at 480 MHz as shown in figure 4-51

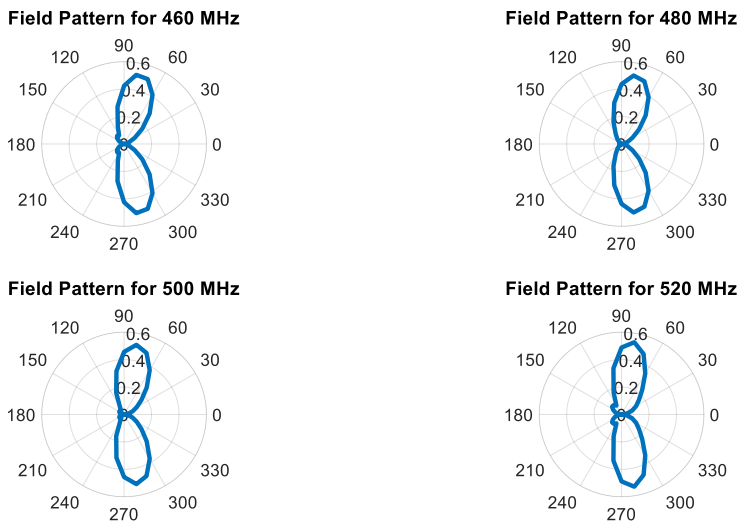


Figure 4-50: Field patterns for 460 – 520 MHz

Figure 4-52 shows that the maximum current distribution for multiple excitation of this monopole antenna is obtained between 520 and 540 MHz. The current distribution values are lower for frequencies below 520 MHz or above 540 MHz.

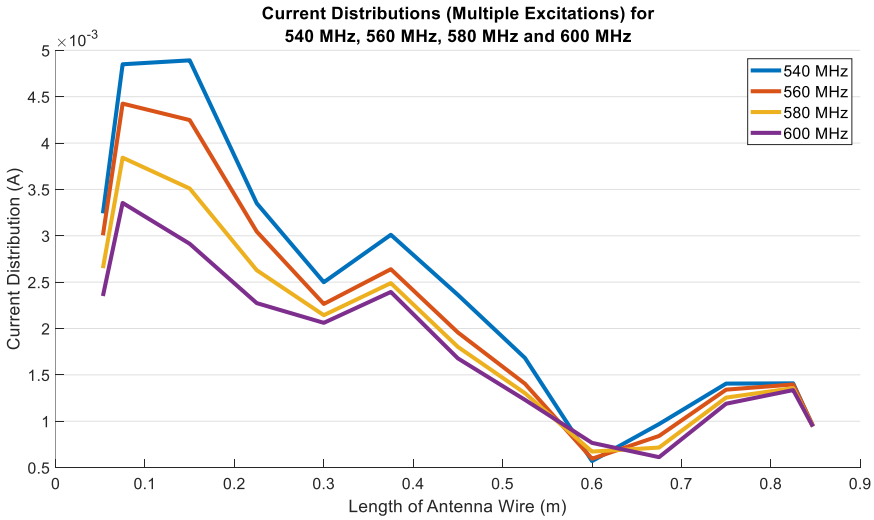


Figure 4-52: Current distributions for 540 - 600 MHz frequencies multiple excitation

Figure 4-53 shows that the propagation distance decreases with increasing frequency from 540 MHz to 600 MHz.

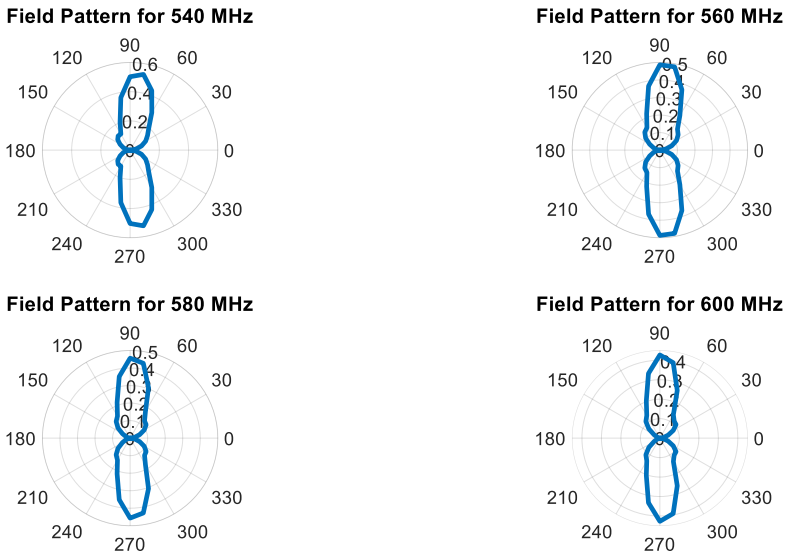
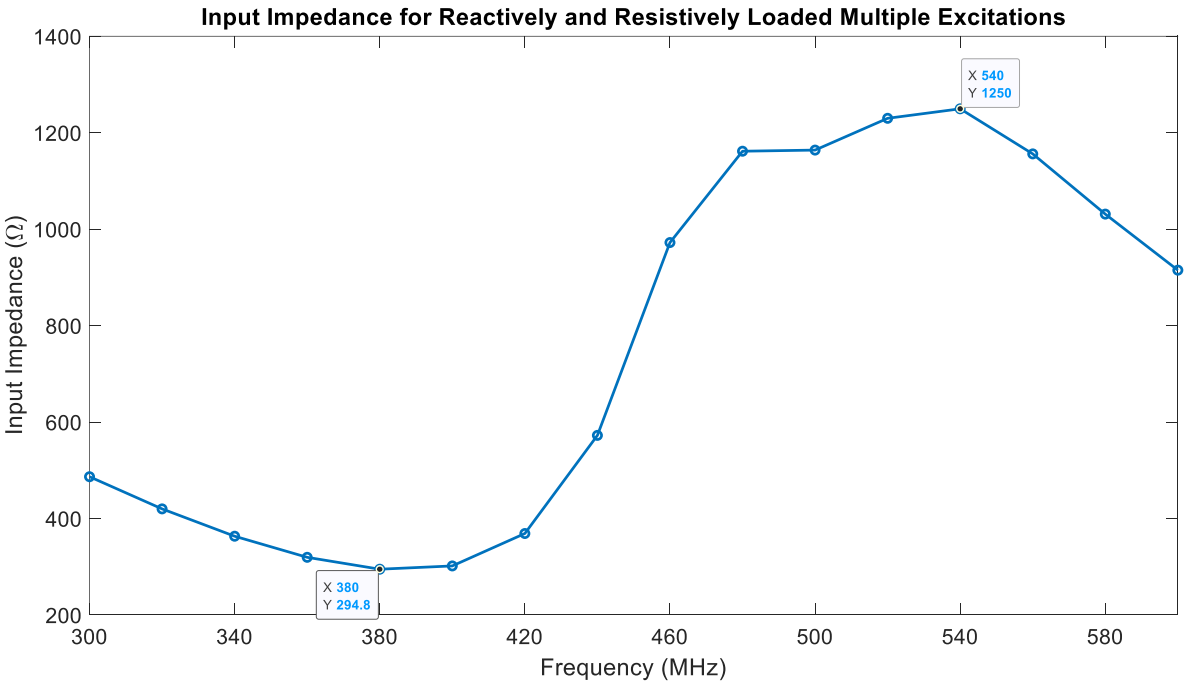


Figure 4-53: Field patterns for 540 – 600 MHz



CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1. Findings

This project was able to use MATLAB scripts to implement method of moments to determine the current distribution, the input impedance and input admittance on a thin wire antenna with both single and multiple excitations with different types of loads: resistive, reactive and combination of both resistive and reactive loads.

It can be seen that the magnitude of the current distribution increases with increasing frequency for the reactive loaded scenario when the thin-wire monopole antenna is singly excited by 1V, feed voltage at 0.436 m of the antenna length for the frequency ranges of 300MHz to 360MHz.

It was found that as the frequency increases, more propagating distances are covered but the angular coverage decreases as represented by the narrower field pattern. The propagation distance is a measure of directivity of the antenna.

The antenna becomes more directional and beamforming with increasing frequencies as indicated by the gradual appearance of side lobes starting from the 400 MHz frequency for the reactive load with single excitation.

The input impedance is a function of frequency and for the radiation problems, the input impedances corresponding to the feedpoints are determined.

The input impedance continues to decrease as the frequency increases from 300 MHz until the minimum value (124.3 Ω) obtained at 480 MHz. Beyond this point, the input impedance steadily rises with increasing frequency.

From the findings, it can then be reccommended that if the antenna design goal is to achieve uniform radiating pattern, then the monopole antenna should be operated between 300 and 360 MHz. If angular coverage of the antenna is the goal with almost equal propagation distance on two halves of the antenna, then the antenna should be operated between 580 and 600 MHz.

However, if maximizing the propagating distance and current distribution magnitude are the goals, then the antenna should be operated between at 480 MHz.

5.2. Conclusion

The problem of determining the current distributions, radiation patterns and input impedances of a thin-wire monopole antenna has existed for many years. Some previously reported works on computing the current distributions, radiation patterns and input impedances of a straight monopole antennas were reviewed.

This dissertation has investigated the analysis of a monopole antenna with single and multiple excitations using method of moments. The monopole antenna with different ranges of frequencies with single and multiple excitations exhibits drastically different charactersitics as the positions or the values of the loadings are varied. A discussion on the results obtained from the analysis was provided.

It can be concluded that the MATLAB program implementation of Method of Moments was effective in determining the current distributions, the input impedance and admittance, and the radiation patterns of a thin-wire monopole antenna for single and multiple excitations and any combination of loads.

5.3. Future Studies

However, this project does not test the MATLAB program developed with multiple points of excitation and multiple points of loading. This can be explored by testing a wire antenna with multi-excitation and different loads at different parts of the wire at the same time.

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Appendix A – Proof of Maxwell's Equations

First Equation - $\nabla \cdot \boldsymbol{D} = \boldsymbol{\rho_v}$

The first Maxwell's equation is based on Gauss electrostatic law which states that a closed surface integral of electric flux is always equal to the charge enclosed over the surface.

$$\oiint \vec{D} \cdot d\bar{s} = Q_{enclosed} \tag{A.1}$$

Since any closed system will have multiple surfaces but a single volume, equation (A.1) can be converted to a volume integral by taking the divergence of the same vector as shown in equation (A.2).

$$\oiint \vec{D} \cdot d\bar{s} = \iiint \nabla \cdot \vec{D} d\vec{v} \tag{A.2}$$

This implies:

$$\iiint \nabla \cdot \vec{D} d\vec{v} = Q_{enclosed} \tag{A.3}$$

Charges in the closed surface ($Q_{enclosed}$) will be distributed over its volume. Therefore, the volume charge density (ρ_v), in C/m^3, can be defined as $\rho_v = \frac{dQ}{dv}$. Therefore, the closed surface charge can be expressed as in equation (A.4). This is the volume charge density over the particular volume

$$Q = \iiint \rho_v dv \tag{A.4}$$

Substituting equation (A.4) into (A.3), we have $\iiint \nabla \cdot \vec{D} d\vec{v} = \iiint \rho_v dv$.

This implies that $\nabla \cdot \vec{D} = \rho_v$

Second Equation - $\nabla \cdot \boldsymbol{B} = \mathbf{0}$

Maxwell's second law is based on Gauss law of magnetostatics which states that a closed surface integral of magnetic flux density is always equal to total scalar magnetic flux enclosed within that surface of any shape or size lying in the medium.

$$\oiint \vec{B}.ds = \phi_{enclosed} \tag{A.5}$$

Scalar magnetic flux (ϕ) is the circular magnetic field generated around a current-carrying conductor while scalar electric flux (ψ) are the imaginary lines of force radiating in an outward direction. Therefore, it can be concluded that magnetic flux cannot be enclosed within a closed

surface of any shape. Therefore, the right-hand side (RHS) of equation (A.5) becomes 0 as shown in equation (A.6).

$$\oiint \vec{B}.ds = 0 \tag{A.6}$$

Converting the surface integral of equation (A.6) to volume integral by taking the divergence of the same vector, we have equation (A.7).

$$\oiint \vec{B}.ds = \iiint \nabla \cdot \vec{B} dv \tag{A.7}$$

This implies $\iiint \nabla \cdot \vec{B} dv = 0$

Since the volume of any body cannot be zero, therefore $\iiint dv \neq 0$. Hence, $\nabla \cdot \vec{B} = 0$.

Where $\vec{B} = \mu\vec{H}$ is the flux density. Therefore, $\nabla \cdot \vec{H} = 0$

Third Equation - $\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial \boldsymbol{t}}$

Maxwell's third equation is derived from Faraday's laws of Electromagnetic induction which states that whenever there are n-turns of conducting coil in a closed path which is placed in a time-varying magnetic field, an alternating electromotive force (emf) gets induced in each coil. According to Lenz's law, an induced emf always opposes the time-varying magnetic flux. Mathematically, the alternating emf (e_a) is expressed in equation (A.8).

$$e_a = -N\frac{d\phi}{dt} \tag{A.8}$$

Where N is the number of turns in a coil; ϕ is the scalar magnetic flux. The negative sign indicates the induced emf always opposes the time-varying magnetic flux. If the number of turns in the coil is one, then equation (A.8) becomes $e_a = -\frac{d\phi}{dt}$. Recall from that equation (A.5) expresses $\phi = \oiint \vec{B}.ds$. Substituting this expression into equation (A.8), we have $e_a = -\frac{d}{dt}\iint \vec{B} \cdot ds$

$$e_a = \iint -\frac{d\vec{B}}{dt} \cdot ds \tag{A.9}$$

The alternating emf (e_a) induced in a coil is a closed path, that is, $e_a = \oint \vec{E} \cdot dl$. Substituting this expression into equation (A.9), we have

$$\oint \vec{E} \cdot dl = \iint -\frac{d\vec{B}}{dt} \cdot ds \tag{A.10}$$

The closed line integral can be converted to surface integral using Stoke's theorem which states that the closed line integral of any vector field is always equal to the surface integral of the curl of the same vector field. Therefore, the left-hand side (LHS) of equation (A.10) can be expressed as:

$$\oint \vec{E} \cdot dl = \iint (\nabla \times \vec{E}) \cdot ds \tag{A.11}$$

Substituting equation (A.11) into (A.10), we have

$$\iint (\nabla \times \vec{E}) \cdot ds = \iint -\frac{d\vec{B}}{dt} \cdot ds \tag{A.11}$$

Hence, $\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$. This implies that the time-varying magnetic field will always produce an electric field. In the case of a static magnetic field, where the static electric field vector is irrotational, the time-varying magnetic field is zero. That is, $-\frac{d\vec{B}}{dt} = 0$. This implies $\nabla \times \vec{E} = 0$.

Fourth Equation - $\nabla \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial \boldsymbol{t}} + \boldsymbol{J}$

Maxwell's fourth equation is based on Ampere's circuit law which states that the closed line integral of the magnetic field vector is always equal to the total amount of scalar electric field enclosed within the path of any shape.

$$\oint \vec{H} \cdot dl = I_{enclosed} \tag{A.12}$$

The LHS of equation (A.12) can be converted to surface integral using Stoke's theorem.

$$\oint \vec{H} \cdot dl = \iint (\nabla \times \vec{H}) \cdot ds \tag{A.13}$$

Substituting equation (A.13) into equation (A.12), we have

$$\iint (\nabla \times \vec{H}) \cdot ds = I_{enclosed} \tag{A.14}$$

Towards the conversion of the scalar quantity $I_{enclosed}$ to vector quantity, we can multiply by the current density vector $\vec{J}$, defined as scalar current flowing per unit surface area. That is, $\vec{J} = \frac{dI}{ds}$. Hence, $I = \iint \vec{J}.ds$. Therefore, equation (A.14) becomes

$$\iint (\nabla \times \vec{H}) \cdot ds = \iint \vec{J}.ds \tag{A.15}$$

Therefore, $\vec{J} = \nabla \times \vec{H}$. This implies that the current density vector is a curl of static magnetic field vector. Applying divergence to equation (A.15), we have

$$\nabla \cdot (\nabla \times \vec{H}) = \nabla \times \vec{J} \tag{A.15}$$

The divergence of the curl of any vector is zero. This implies $\nabla \cdot (\nabla \times \vec{H}) = 0$. Applying the time-varying field, we have $\nabla \times \vec{J} = \frac{\partial \rho_v}{\partial t}$. Therefore, $\frac{\partial \rho_v}{\partial t} = 0$. However, this contradicts the equation of time-varying fields. In the interest of overcoming this limitation, let $\vec{G}$ be added to the current density ($\vec{J}$) of the expression $\vec{J} = \nabla \times \vec{H}$ such that:

$$\nabla \times \vec{H} = \vec{J} + \vec{G} \tag{A.16}$$

Now, taking the divergence of both sides of equation (A.16), we have $\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot (\vec{J} + \vec{G})$. As earlier stated, the divergence of any curl is zero, therefore, $\nabla \cdot \vec{J} + \nabla \cdot \vec{G} = 0$

$$\nabla \cdot \vec{G} = -\nabla \cdot \vec{J} \tag{A.17}$$

Recall that $\nabla \times \vec{J} = \frac{\partial \rho_v}{\partial t}$, therefore $\nabla \cdot \vec{G} = \frac{\partial \rho_v}{\partial t}$. Also, recall from Maxwell's first equation that $\rho_v = \nabla \cdot \vec{D}$. Therefore,

$$\nabla \cdot \vec{G} = \frac{\partial (\nabla \cdot \vec{D})}{\partial t} \tag{A.18}$$

The $\frac{\partial}{\partial t}$ of equation (A.18) is time-variant while $\nabla \cdot \vec{D}$ is space-variant and are bot independent of each other. Equation (A.18) can be rearranged such as $\nabla \cdot \vec{G} = \nabla \cdot \frac{\partial \vec{D}}{\partial t}$. Therefore,

$$\vec{G} = \frac{\partial \vec{D}}{\partial t} \tag{A.19}$$

Substituting equation (A.19) into (A.16), we have Maxwell's fourth equation as shown in equation (A.20)

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \tag{A.20}$$

$\frac{\partial \vec{D}}{\partial t}$ is known as displacement current density $(\vec{J}_D)$

It is known that the magnetic flux is equal to the product of the electric flux and permittivity. hence, $\vec{D} = \varepsilon \vec{E}$.

Therefore, $\frac{\partial \vec{D}}{\partial t} = \frac{\partial(\varepsilon \vec{E})}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t}$. Hence, the displacement current density $(\vec{J}_D)$ can also be stated as $\vec{J}_D = \varepsilon \frac{\partial \vec{E}}{\partial t}$.

Appendix B – MATLAB Code

Radiation Script

```
%This program calculates for a z-directed wire antenna

clc;
close all;
NW=input('Enter the No of wires: ');
NP=input('Enter the No of equally spaced points: ');
NR=input('Enter the No of wires radii encountered: ');
S1=1/((NP-1)*2);
% New Inputs Begin
freq = 300e6;    %300e6:20e6:600e6;
spe = 3e8;
wavl = spe./freq;
bk=(2*pi)./wavl;    %6.283184;
PX=zeros(1,NP);
PY=zeros(1,NP);
PZ=zeros(1,NP);
IFP=zeros(1,20);
for I = 1:25
      XI=I;
      PX(I)=0;
      PY(I)=0;
      PZ(I)= (XI-13)*S1;
end
L=zeros(1,NW+1);
LL=zeros(1,NW+1);
for P=1:NW
    disp(['For wire ', num2str(P)]);
    LL(P)= input('Enter the No of the first point : ');
end
LR=zeros(1,NR+1);
for P=1:NW
    disp(['For wire ', num2str(P)]);
    LR(P)= input('Enter the No of the point where radius changes : ');
end
rad=0.00337.*wavl;  %;0.01.*wavl;
RAD2=rad.^2;
fprintf('\n\n ************************************\n\n');
fprintf('Number of wires: %1.0f\n', NW);
fprintf('Number of points: %1.0f\n', NP);
fprintf('Number of radii encountered: %1.0f\n', NR);
% f1 = figure('Name','Current Distribution','NumberTitle','off');

for wavL = 1:length(wavl)
    J1=1;
    J2=2;
    N1=0;
    fprintf('2 * pi/wavelength: %1.6f\n', bk(wavL));
    fprintf('radius of the wire: %1.6f\n', rad(wavL));
    BK = bk(wavL);
    RAD = rad(wavL);
    for J=1:NP
        if (LL(J1)-J)==0
            J2=J2-1;
            L(J1)=J2;
            J1=J1+1;
        else
            N1=N1+1;
            J3=J-1;
```

```
            FN=N1/2;
            fn=fix(FN);
            if fn==N1/2
                J2=J2+1;
            end
            XX(N1)=0.5*(PX(J)+PX(J3));
            XY(N1)=0.5*(PY(J)+PY(J3));
            XZ(N1)=0.5*(PZ(J)+PZ(J3));
            SS1=PX(J)-PX(J3);
            SS2=PY(J)-PY(J3);
            SS3=PZ(J)-PZ(J3);
            SS4=sqrt((SS1*SS1)+(SS2*SS2)+(SS3*SS3));
            TX(N1)=SS1/SS4;
            TY(N1)=SS2/SS4;
            TZ(N1)=SS3/SS4;
            AL(N1)=SS4;
        end
    end
    L(J1)=J2;
    N=J2-2;
    calz; % Calculates the impedance matrix, Z
    NL=1;
    fprintf('Number of loads: %1.0f\n', NL);
    LP=6; %[3 12]
    fprintf('Load Position: %1.0f\n', LP);
    ZL=100.0+100.00i; %[0-65.97i 0-65.97i]; %;
    fprintf('Corresponding load: %4.2f + j%4.2f\n', real(ZL),imag(ZL));
    J2=(LP-1)*N+LP;
    Z(J2)=Z(J2)+ZL;
    lineq; % Inverts the impedance matrix i.e 1/Z
    for I= 1:N
        U(I)=0.;
    end
    NF=1;   %number of independent feed voltages
    J1=3; %number of the triangle of the first excitation voltage
    V=1.0+0.0i; %[0.525-0.340i 1.017-0.524i 0.525-0.340i];
    U(J1)=V(1);
    IFP = 3; %[3 7 12]; %numbers of triangles where excitations are applied
%     for I=1:NF
%         J1=3; %number of the triangle of the first excitation voltage
%         V=[0.525-0.340i 1.017-0.524i 0.525-0.340i]; %1.0+0.0i;
%         U(J1)=V(1);
%         IFP(I)=J1;
%     end
    fprintf('Number of feed Point: %1.0f\n', NF);
    fprintf('Feed Position: %1.0f\n',J1);
    fprintf('Corresponding Feed Voltage: %1.2f\n\n\n', V);
    J3=0;
    J1=0;
    J1=J1+1;
    U1=0.;
    J2=0;
    J2=J2+1;
    J3=J3+1;
    U1=U1+Z(J3)*U(J2);
    while (J2-N)<0
        J2=J2+1;
        J3=J3+1;
        U1=U1+Z(J3)*U(J2);
    end
    C(J1)=U1;
```

```
    while (J1-N)<0
        J1=J1+1;
        U1=0.;
        J2=0;
        J2=J2+1;
        J3=J3+1;
        U1=U1+Z(J3)*U(J2);
        while (J2-N)<0
            J2=J2+1;
            J3=J3+1;
            U1=U1+Z(J3)*U(J2);
        end
        C(J1)=U1;
    end
    fprintf('CURRENT DISTRIBUTION\n');
    fprintf(' I\t Length\t Real\t\t\t Imaginary\t\t\t Magnitude\t\t
Phase\n');
    UPCM = zeros(1,N);
    leng=(2*SS3):(2*SS3):(N*2*SS3);
    for I=1:N
        UP=C(I);% i changed U1 to an array UP.
        CM=abs(UP);
        if (CM)~=0
            CP=atan2(imag(UP),real(UP))*57.2858;
        else
            CP=0.;
        end
        fprintf('%2.0f\t %6.4f\t %12.6e\t %12.6e\t\t %12.6e\t %7.3f\n',
I,leng(I),real(UP),imag(UP),CM,CP);
        UPCM(I) = CM;
    end
    % conversion of triangular functions to antenna wire length

%     hold on
    plot(leng,UPCM,'LineWidth',4)
    xlabel('Length of Antenna Wire (m)','FontSize',14)
    ylabel('Current Distribution (A)','FontSize',14)
%     title(['Current Distribution for ',num2str(freq(wavL)/1e6), '
MHz'],'FontSize',16);
%     title('Current Distribution for Resistive and Reactive Load',
'FontSize',16);
    POWER=0;
    for i=1:NF
        J1=IFP(i);
        ZIN=U(J1)/C(J1);
        fprintf('\nInput Impedance at position 3 = %11.6e
+j%11.6e\n',real(ZIN),imag(ZIN));
        GM=abs(ZIN);
        if (GM-0.1e-6)>0
            YIN=1./ZIN;
        end
        fprintf('\nInput Admitance at position 3 = %11.6e
+j%11.6e\n',real(YIN),imag(YIN));
        S1=real(ZIN);
        S2=abs(C(J1));
        POWER=POWER+0.5*S2*S2*S1;
    end
    fprintf('\n Total Power Input = %11.6e\n',POWER);
    %FIELD PATTERN CALCULATIONS
    fprintf('\n FIELD PATTERN\n K=1 for thetha component, K=2 for phi
component\n')
```

```
    fprintf('\n Theta\t phi\t k\t\t Real\t\t imaginary\t\t\t Magnitude\t\t
Phase\n')
    PHI=90.;
    PH=PHI*0.0174533;
    k=1;
    UTGM = [];
    theta = [];
    for ITH=1:10:361
        THE=ITH-1;
        TH=THE*0.0174533;
        row;
        for K=1:2
            %UT(k)=E(K);
            UT=E(K);
            %GM(k)=abs(UT(k));
            GM=abs(UT);
            %if (GM(k))>0
            if (GM)>0
                %EP(k)=atan2(imag(UT(k)),real(UT(k)));
                EP=atan2(imag(UT),real(UT));
            else
                %EP(k)=0.;
                EP=0.;
            end
            %GP(k)=EP(k)*57.2858;
            GP=EP*57.2858;
            k=k+1;
            fprintf('\t%2.0f\t %2.0f\t\t %1.0f\t %11.6e\t %11.6e\t\t
%11.6e\t %7.3f\n',THE,PHI,K,real(UT),imag(UT),GM,GP);
            if K == 1
                theta = [theta THE*pi/180];
                UTGM = [UTGM GM];
            end
        end
    end
    figure('Name',['Field Pattern for ',num2str(freq(wavL)/1e6), '
MHz'],...
        'NumberTitle','off');
    polarplot(theta,UTGM,'LineWidth',4)
    title(['Field Pattern for ',num2str(freq(wavL)/1e6), '
MHz'],'FontSize',16);
    for M=1:3
        XM=M;
        for K=1:2:7
            XK=K;
            X1=0;
            X2=0.2*XK;
            X3=(XM-1)*0.5;
            enear;
            fprintf('\n X = %4.4f\t Y = %4.4f\t Z = %4.4f\n',X1,X2,X3);
            fprintf('\t J\t\t\t\t E(J)\t\t\t\t\t Magnitude\t\t Phase\n');
            for J=1:3
                u1=E(J);
                gm=abs(u1);
                if(gm)>0
                    ep=atan2(imag(u1),real(u1));
                else
                    ep=0.;
                end
                gp=ep*57.2858;
```

```
                fprintf('\t%1.0f\t %11.6e\t %11.6e\t %11.6e\t
%6.3f\n',J,real(u1),imag(u1),gm,gp);
            end
        end
    end
end
```

Function ROW

```
% Calculates the far-field pattern
U=0.0+1.0i; %[0.525-0.340i 1.017-0.524i 0.525-0.340i];
eta=376.7307;
CT= cos(TH);
ST= sin(TH);
cp= cos(PH);
SP= sin(PH);
S1=CT*cp;
S2=CT*SP;
BK1=BK*ST*cp;
BK2=BK*ST*SP;
BK3=BK*CT;
N2=1;
N3=-2;
for NS=1:N
    if (L(N2)-NS)==0
        KK=1;
        N3=N3+2;
        N2=N2+1;
    else
        KK=3;
    end
    for K=KK:4
        N7=N3+K;
        BKR(N7)=XX(N7)*BK1+XY(N7)*BK2+XZ(N7)*BK3;
        DT(N7)=TX(N7)*S1+TY(N7)*S2-TZ(N7)*ST;
        DP(N7)=-TX(N7)*SP+TY(N7)*cp;
    end
    N3=N3+2;
end
N2=1;
N3=-2;
U3=0.;
U4=0.;
for NS=1:N
    if (L(N2)-NS)==0
        N3=N3+2;
        N2=N2+1;
    end
    J1=(NS-1)*4;
    U1=0.;
    U2=0.;
    for JS=1:4
        J2=J1+JS;
        J3=N3+JS;
        S1=BKR(J3);
        U5=T(J2)*(cos(S1)+U*sin(S1));
        U1=U1+U5*DT(J3);
        U2=U2+U5*DP(J3);
    end
    U3=U3+U1*C(NS);
    U4=U4+U2*C(NS);
```

```
    N3=N3+2;
end
S1=0.0795774*eta*BK;
E(1)=-U*S1*U3;
E(2)=-U*S1*U4;
```

Function Lineq

```
%This fuction inverts the impedance matrix to generate the admittance
%matrix
LA=zeros(1,20);
for I=1:N
    LA(I)=I;
end
M1=0;
for M=1:N
    K=M;
    for I=M:N
        K1=M1+I;
        K2=M1+K;
        if(abs(Z(K1))-abs(Z(K2)))>0 % change from while to if
            K=I;
        end
    end
    LS=LA(M);
    LA(M)=LA(K);
    LA(K)=LS;
    K2=M1+K;
    STOR=Z(K2);
    J1=0;
    for J=1:N
        K1=J1+K;
        K2=J1+M;
        STO=Z(K1);
        Z(K1)=Z(K2);
        Z(K2)=STO/STOR;
        J1=J1+N;
    end
    K1=M1+M;
    Z(K1)=1./STOR;
    for I=1:N
        if (I-M)~=0 % i change this from while to if.
            K1=M1+I;
            ST=Z(K1);
            Z(K1)=0.+0.i;
            J1=0;
            for J=1:N
                K2=J1+M;
                K1=J1+I;
                Z(K1)=Z(K1)-Z(K2)*ST;
                J1=J1+N;
            end
        end
    end
    M1=M1+N;
end
J1=0;
for J=1:N
    while (J-LA(J))~=0
        LAJ=LA(J);
        J2=(LAJ-1)*N;
```

```
        for I=1:N
            K2=J2+I;
            K1=J1+I;
            S=Z(K2);
            Z(K2)=Z(K1);
            Z(K1)=S;
        end
        LA(J)=LA(LAJ);
        LA(LAJ)=LAJ;
    end
    J1=J1+N;
end
```

Function ENEAR

```
P=zeros(1,50);
U=0.0+1.0i;%[0.525-0.340i 1.017-0.524i 0.525-0.340i];
eta=376.7307;
C1=0.125/pi;
C2=0.25/pi;
U4=-U*BK*eta;
U5=-U*eta/BK;
BK2=BK*BK/2;
BK3=BK2*BK/3;
BK4=BK3*BK/4;
N2=1;
N3=-2;
for NS=1:N
    if (L(N2)-NS)==0
        KK=1;
        N3=N3+2;
        N2=N2+1;
    else
        KK=3;
        N4=N2-1;     % added
    end
    for K=KK:4
        N7=N3+K;
        S1=XX(N7)-X1;
        S2=XY(N7)-X2;
        S3=XZ(N7)-X3;
        R2=S1^2+S2^2+S3^2;
        R=sqrt(R2);
        DX(N7)=S1/R;
        DY(N7)=S2/R;
        DZ(N7)=S3/R;
        RT=(S1*TX(N7))+(S2*TY(N7))+(S3*TZ(N7));
        RT2=RT^2;
        RH=R2-RT2;
        ALP=0.5*AL(N7);
        AR=ALP/R;
        S1=BK*R;
        U2=cos(S1)-U*sin(S1);
        if (AR-0.1)>0            % changed + to -
            U2=U2*C1/ALP;
            U3=U2;
            S1=RT-ALP;
            S2=RT+ALP;
            S3=sqrt(S1^2+RH);
            S4=sqrt(S2^2+RH);
            if (S1)<=0
```

```
                AI1=10^((S2+S4)*(-S1+S3)/RH); % changed log to 10^
            else
                AI1=10^((S2+S4)/(S1+S3));    % changed log to 10^
            end
            AI2=AL(N7);
            AI3=(S2*S4-S1*S3+RH*AI1)/2.;
            AI4=AI2*(RH+ALP*ALP/3+RT2);
            S3=AI1*R;
            S1=AI1-BK2*(AI3-R*(2*AI2-S3));
            S2=-BK*(AI2-S3)+BK3*(AI4-3*AI3*R+R2*(3*AI2-S3));
            RHH=sqrt(abs(RH));
            S3=4*RHH*AI2+(4*R2-AI2^2); % . or +; not changed
            AI0=atand(S3)/RHH;
            S5=AI0*R;
            S3=AI0-BK2*(AI2-R*(2*AI1-S5))+BK4*(AI4-4*R*AI3+6*R2*AI2-...
                R2*R*(4*AI1-S5)); % Removed R2
            S4=-BK*(AI1-S5)+BK3*(AI3-3*R*AI2+R2*(3*AI1-S5));
        else
            U2=U2*C2/R;
            U3=U2/R;
            BA=BK*ALP;
            BA2=BA^2;
            AR2=AR^2;
            AR3=AR2*AR;
            AR4=AR3*AR;
            ZR=RT/R;
            ZR2=ZR^2;
            ZR3=ZR2*ZR;
            ZR4=ZR3*ZR;
            H1=(3-30*ZR2+35*ZR4)*AR3/40.;
            A1=AR*(-1+3*ZR2)/6.;
            A0=1+AR*(A1+H1);
            A1=A1+H1;
            A2=-ZR2/6-AR2*(1-12*ZR2+15*ZR4)/40.;
            A3=AR*(3*ZR2-5*ZR4)/60.;
            A4=ZR4/120;
            S1=A0+BA2*(A2+BA2*A4);
            S2=BA*(A1+BA2*A3);
            A0=1+AR2*(-1+4*ZR2)/3+AR4*(1-12*ZR2+16*ZR4)/5;
            A1=AR*(-1+5*ZR2)/6+AR3*(5-66*ZR2+93*ZR4)/40;
            A2=-ZR2/6-AR2*(1-18*ZR2+29*ZR4)/40;
            A3=AR*(3*ZR2-7*ZR4)/60;
            S3=A0+BA2*(A2+BA2*A4);
            S4=BA*(A1+BA2*A3);
        end
        P(N7)=U2*(S1+U*S2);
        PS(N7)=U3*(S3+U*S4)+U*BK*P(N7);
    end
    N3=N3+2;
end
E(1)=0;
E(2)=0;
E(3)=0;
N2=1;
N3=-2;
for NS=1:N
    if(L(N2)-NS)==0
        N3=N3+2;
        N2=N2+1;
    end
    J1=(NS-1)*4;
```

```
    U1=0;
    U2=0;
    U3=0;
    UA=0;
    UB=0;
    UC=0;
    for JS=1:4
        J2=J1+JS;
        J3=N3+JS;
        UU=T(J2)*P(J3);
        U1=U1+UU*TX(J3);
        U2=U2+UU*TY(J3);
        U3=U3+UU*TZ(J3);
        UU=TP(J2)*PS(J3);
        UA=UA+UU*DX(J3);
        UB=UB+UU*DY(J3);
        UC=UC+UU*DZ(J3);
    end
    N3=N3+2;
    E(1)=E(1)+(U4*U1+U5*UA)*C(NS);
    E(2)=E(2)+(U4*U2+U5*UB)*C(NS);
    E(3)=E(3)+(U4*U3+U5*UC)*C(NS);
end
```

function calz

```
% This function calculates the impedance matrix
Z=zeros(1,400);
U=0.0+1.0i; %[0.525-0.340i 1.017-0.524i 0.525-0.340i];
PSI=zeros(1,200);
DC=zeros(1,200);
T=zeros(1,50);
TP=zeros(1,50);
eta=376.7307;
C1=0.125/pi;
C2=0.25/pi;
J1=1;
J2=-2;
for J=1:N
       if(L(J1)-J)==0
            J2=J2+2;
            J1=J1+1;
       end
       J3=(J-1)*4;
       J4=J3+1;
       J5=J4+1;
       J6=J5+1;
       J7=J6+1;
       K4=J2+1;
       K5=K4+1;
       K6=K5+1;
       K7=K6+1;
       s1= AL(K4)+AL(K5);
       s2=AL(K6)+AL(K7);
       T(J4)=AL(K4)*0.5*AL(K4)/s1;
       T(J5)=AL(K5)*(AL(K4)+0.5*AL(K5))/s1;
       T(J6)=AL(K6)*(AL(K7)+0.5*AL(K6))/s2;
       T(J7)=AL(K7)*0.5*AL(K7)/s2;
       TP(J4)=AL(K4)/s1;
       TP(J5)=AL(K5)/s1;
       TP(J6)=-AL(K6)/s2;
```

```
        TP(J7)=-AL(K7)/s2;
        J2=J2+2;
end
U3= U*BK*eta;
U4= -U/BK*eta;
BK2=BK*BK/2.;
BK3=BK2*BK/3.;
N9=0;
N2=1;
N0=1;
N3=-2;
for NS=1:N
    if (L(N2)-NS)==0
        KK=1;
        N3=N3+2;
        N2=N2+1;
    else KK=3;
        for NF=1:N1
            N4=NF+N1;
            N5=N4+N1;
            N6=N5+N1;
            DC(NF)=DC(N5);
            DC(N4)=DC(N6);
            PSI(NF)=PSI(N5);
            PSI(N4)=PSI(N6);
        end
    end
    if(N3+1-LR(N0))==0
        AA=RAD2(N0);
        N0=N0+1;
    end
    for K=KK:4
        N7=N3+K;
        K1=(K-1)*N1;
        for NF=1:N1
            N8=NF+K1;
            S1=XX(N7)-XX(NF);
            S2=XY(N7)-XY(NF);
            S3=XZ(N7)-XZ(NF);
            R2=(S1*S1)+(S2*S2)+(S3*S3)+AA;
            R=sqrt(R2);
            RT=abs((S1*TX(N7))+(S2*TY(N7))+(S3*TZ(N7)));
            RT2=RT*RT;
            RH=(R2-RT2);
            ALP=0.5*AL(N7);
            AR=ALP/R;
            S1=BK*R;
            U2=cos(S1)-U*sin(S1);
            if (AR-0.1)>0
                U2=U2*C1/ALP;
                S1=RT-ALP;
                S2=RT+ALP;
                S3=sqrt(S1*S1+RH);
                S4=sqrt(S2*S2+RH);
                if S1<=0
                    AI1=log((S2+S4)*(-S1+S3)/RH);
                else AI1=log((S2+S4)/(S1+S3));
                end
                AI2=AL(N7);
                AI3=((S2*S4)-(S1*S3)+(RH*AI1))/2.;
                AI4=AI2*(RH+(ALP*ALP/3)+RT2);
```

```
                S3=AI1*R;
                S1=AI1-BK2*(AI3-R*(2.*AI2-S3));
                S2=-BK*(AI2-S3)+BK3*(AI4-3.*AI3*R+R2*(3.*AI2-S3));
            else
                U2=U2*C2/R;
                BA=BK*ALP;
                BA2=BA*BA;
                AR2=AR*AR;
                AR3=AR2*AR;
                ZR=RT/R;
                ZR2=ZR*ZR;
                ZR3=ZR2*ZR;
                ZR4=ZR3*ZR;
                H1=(3.*-30.*ZR2+35.*ZR4)*AR3/40.;
                A1=AR*(-1.+3.*ZR2)/6.+(3.-30.*ZR2+35.*ZR4)*AR3/40.;
                A0=1.+AR*A1;
                A2=-ZR2/6.-AR2*(1.-12.*ZR2+15.*ZR4)/40.;
                A3=AR*(3.*ZR2-5.*ZR4)/60.;
                A4=ZR4/120.;
                S1=A0+BA2*(A2+BA2*A4);
                S2=BA*(A1+BA2*A3);
            end
            PSI(N8:N8+2)=U2.*(S1+U.*S2);
            DC(N8)=(TX(NF)*TX(N7))+(TY(NF)*TY(N7))+(TZ(NF)*TZ(N7));
        end
    end
    N3=N3+2;
    J3=(NS-1)*4;
    J7=-2;
    J9=1;
    for NF=1:N
        J1=(NF-1)*4;
        if (L(J9)-NF)==0
            J9=J9+1;
            J7=J7+2;
        end
        N9=N9+1;
        U5=0.;
        U6=0.;
        J5=0;
        for JS=1:4
            J4=J3+JS;
            J8=J5+J7;
            for JF=1:4
                J6=J8+JF;
                J2=J1+JF;
                U5=T(J2)*T(J4)*DC(J6)*PSI(J6)+U5;
                U6=TP(J2)*TP(J4)*PSI(J6)+U6;
            end
            J5=J5+N1;
        end
        Z(N9)=U5*U3+U6*U4;
        J7=J7+2;
    end
end
```